

Abdul Wahid

EC-Council CND (Certified Network Defender)

Linux System Hardening & Security Operations – Automated Monitoring & Maintenance

Executive Summary

Module 6 focuses on Linux System Hardening & Security Operations. It builds a segmented lab network protected by a Smoothwall firewall (GREEN internal / RED external) and applies layered hardening controls on an Ubuntu "Operation Dept" VM.

Key activities include:

- Network isolation with Smoothwall and static IP configuration via Netplan
- Linux security best practices (disable unnecessary services, SSH hardening, user/group management, UFW firewall, key-based authentication)
- Strong password policies with pwquality and login.defs
- Security auditing with Lynis and rootkit scanning with rkhunter
- File integrity monitoring using SHA-256 checksums
- Malware detection with ClamAV
- Kernel-level event auditing with AuditD
- Kernel management and system cleanup

All major controls are automated via reusable scripts and scheduled with cron (daily integrity/audit checks, weekly malware scans and cleanup). The module emphasizes defense-in-depth, continuous monitoring, and a validate-after-configure approach, resulting in a measurably more secure and sustainable Linux environment.

Contents

1. Virtual Network Editor Setting 7
2. Smoothwall Firewall Network Setup 8
2.1 VM Network Adapter Configuration 8
2.2 Smoothwall Installation / Configuration Steps (text console) 9
2.3 Verify Network with ip addr 14
3. Ubuntu "Operation Dept" VM — Network Setup 15
3.1 Configure Static IP via Netplan 15
3.2 Apply and Verify Connectivity 16
3.3 Check Connectivity to Domain Controller (DC01) 16
LAB 1 — Linux Security Best Practices 17
Executive Summary 17
1.1 List Running Services 17
1.2 Disable Unnecessary Services 18
Disable Apache2 (if it exists): 18
Disable avahi-daemon: 18
1.3 SSH Hardening — Restrict Root Login & Disable X11 Forwarding 19
1.4 Create a New User and Group 20
1.5 Password Aging & Expiry Policy 21
1.6 SSH Access Policy Rules 22
1.7 Check Config Syntax and Restart SSH 22
1.8 UFW Firewall Setup 24
1.9 SSH Permissions Hardening 25
1.10 Sudo Access for Abdul 25
1.11 Create SSH Directory & Generate SSH Key for Abdul 27

1.12 Test Login via PuTTY 28
LAB 2 — Implementation of Password Policies 30
Executive summary 30
2.1 Install pwquality 30
2.2 Policy Check — Try a Weak Password 30
2.3 Login Defense Configuration (/etc/login.defs) 31
2.4 Sudoers Timeout Settings 31
2.5 Verify All Password Policies 32
LAB 3 — Installing Lynis Security Auditing 33
Executive summary 33
3.1 Install and Check Version 33
3.2 Run a Full System Audit 34
3.3 View the Security Audit Report 35
3.4 Cleaner Re-run and Log Review 35
3.5 Install a Malware Scanner (rkhunter) 36
3.6 Run Initial rkhunter Scan 38
3.7 View Final Rootkit/Malware Report 39
3.8 Re-run Lynis to Confirm Improved Hardening Score 40
LAB 4 — File Integrity Checking with Checksums (SHA-256) 41
Executive summary 41
4.1 Create Test Files 41
4.2 Locate sha256sum 41
4.3 Generate Checksums 41
4.4 Verify Checksum File Content 42
4.5 Check File Integrity 42
4.6 Demonstrate Integrity Failure After Modification 42
4.7 Create Two More Files 43

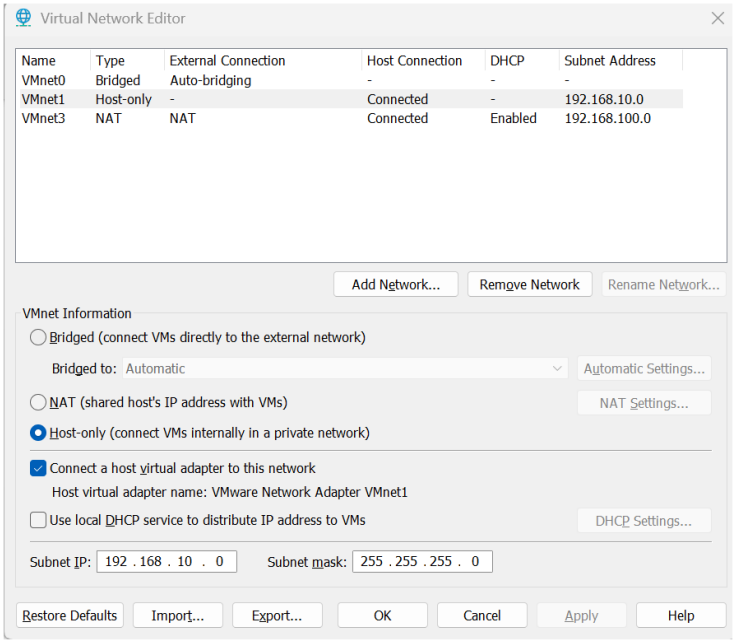
4.8 Checksum Critical System Files 43
4.9 Create an Automated Integrity Check Script 44
4.10 Make Executable and Test 45
4.11 Schedule Daily Automated Check via Cron....45
LAB 5 — Malware Scanning with ClamAV 47
Executive summary....47
5.1 Install ClamAV and ClamTK 47
5.2 Start and Enable ClamAV Daemon 47
5.3 Stop Clam, Fix Lock File and Log Directory Permissions 48
5.4 Update the Virus Database 48
5.5 Restart ClamAV Daemon....49
5.6 Create Test Files and Download EICAR Test File 49
5.7 Scan the Test Directory....50
5.8 Scan the User's Home Directory 50
5.9 Create an Automated Weekly Malware Scan Script 51
5.10 Make Executable 51
5.11 Add to Crontab 52
5.12 Test the Script Manually....53
LAB 6 — File Monitoring and Event Auditing with AuditD....54
Executive summary....54
6.1 Install AuditD....54
6.2 Start and Enable AuditD 55
6.3 Add Audit Rules for File Monitoring 55
6.4 Trigger a Test Audit Event 57
6.5 Search Audit Logs for Changes....58
6.6 View Detailed Raw Audit Logs 58
6.7 Generate a Summary Audit Report....59

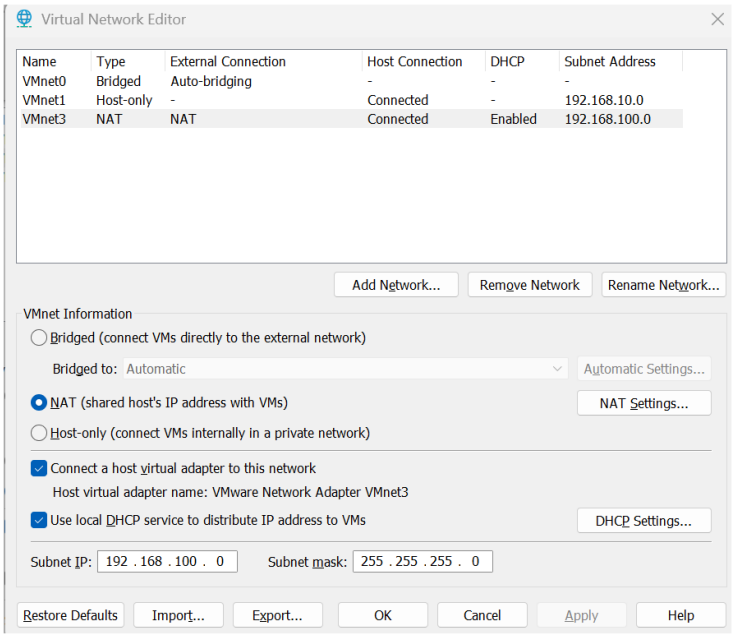
6.8 Create an Audit Monitoring Report Script 60
6.9 Make Executable and Verify by Running.....61
6.10 Add to Crontab 61
LAB 7 — Kernel Management & System Cleanup.....63
Executive summary.....63
7.1 Check Current Running Kernel.....63
7.2 List All Installed Kernels.....63
7.3 List All Kernel-Related Packages 64
7.4 List Installed Kernel Versions 64
7.5 Check Boot Partition Usage 64
7.6 List All Installed Linux Packages (Full).....65
7.7 Remove Old/Unused Kernel Packages 65
7.8 Verify Cleanup 67
7.9 Fix Broken Dependencies 68
7.10 Remove Broken Libraries / Final Cleanup Pass.....68
7.11 Verify System Health and Disk Space.....69
7.12 Create an Automated System Cleanup Script 70
7.13 Make Executable and Test 71
7.14 Schedule Weekly Automated Cleanup via Cron 72
Conclusion.....74

1. Virtual Network Editor Setting

The VMware Virtual Network Editor is used to configure the virtual networks (VMnets) used by the lab VMs.

- **VMnet1** — Host-only network, Subnet: 192.168.10.0, connected to host, no DHCP.
- **VMnet3** — NAT network, Subnet: 192.168.100.0, DHCP enabled, connected to host.



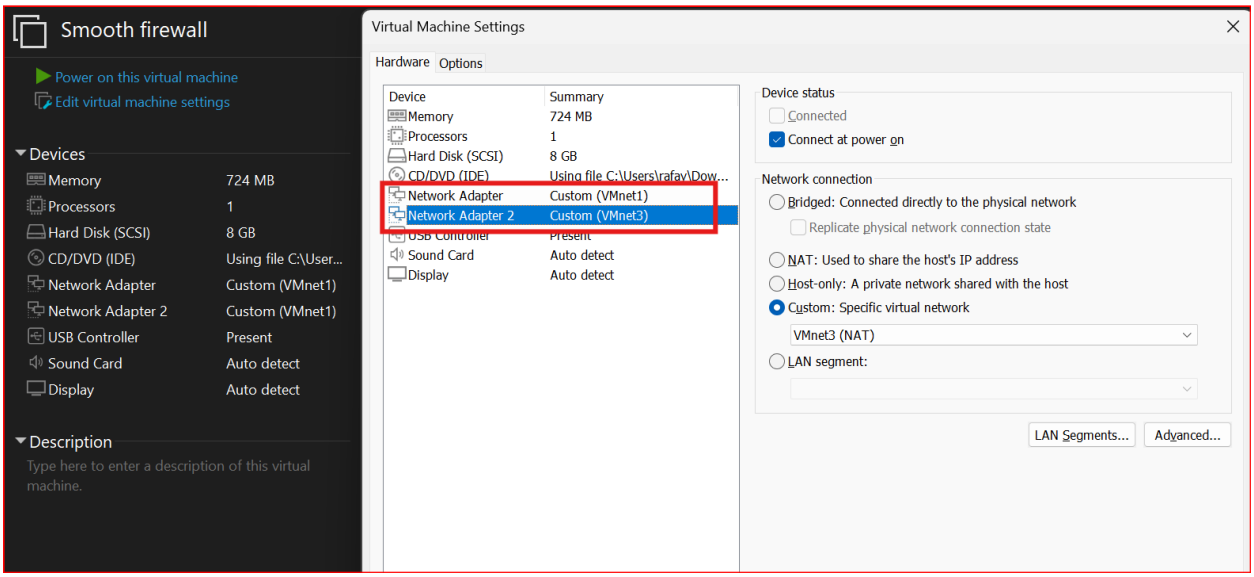


2. Smoothwall Firewall Network Setup

2.1 VM Network Adapter Configuration

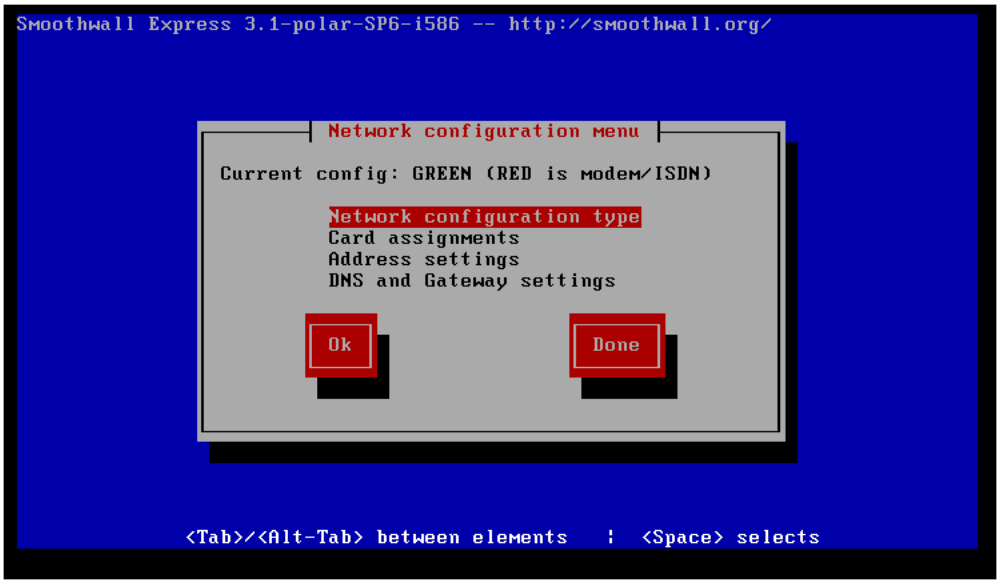
The Smoothwall firewall VM has two network adapters:

- **Network Adapter 1** → Custom (VMnet1) — connects to the internal/GREEN network
- **Network Adapter 2** → Custom (VMnet3) — connects to the NAT/RED network

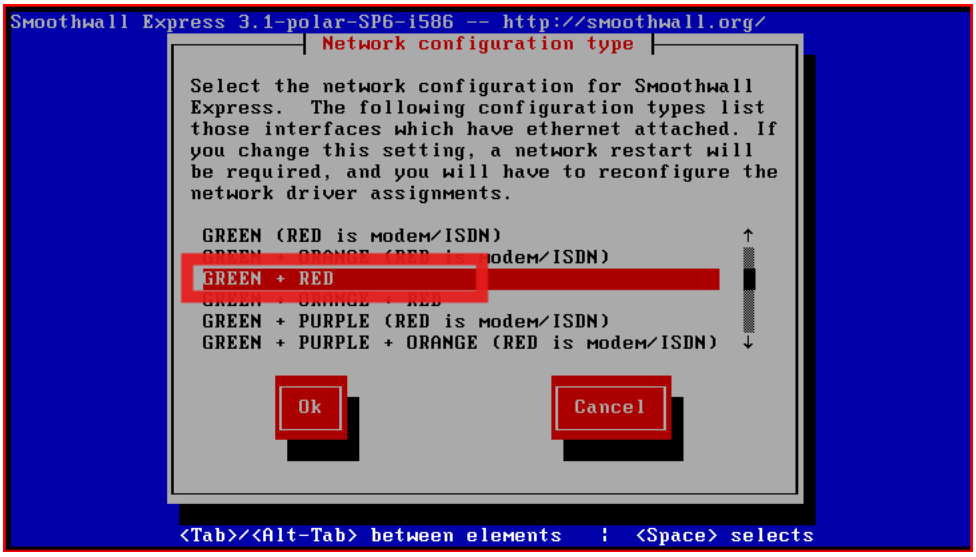


2.2 Smoothwall Installation / Configuration Steps (text console)

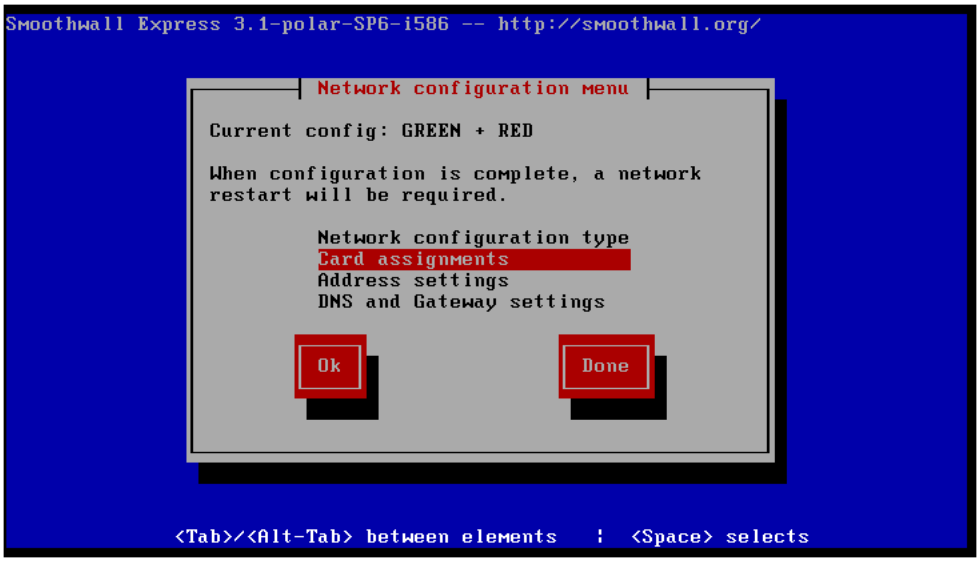
1. Boot Smoothwall Express 3.1. Open **Network configuration menu → Network configuration type**.



2. Select configuration type: **GREEN + RED**.

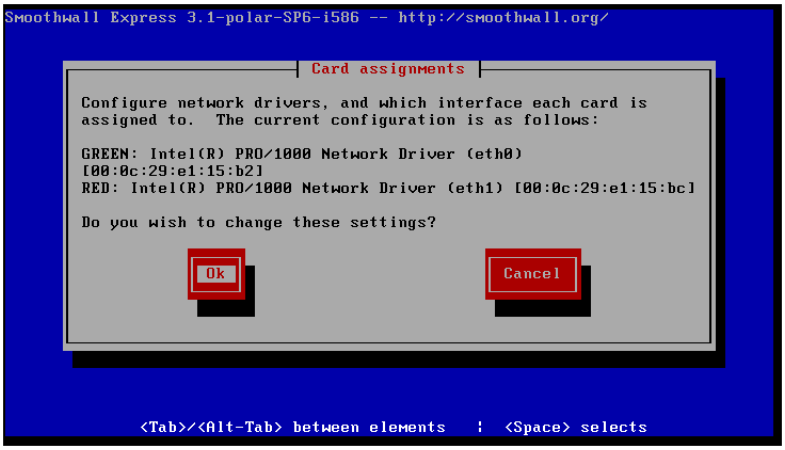


3. Go to **Card assignments** — the system detects an unclaimed Ethernet card (Intel PRO/1000). Assign it to **GREEN**.

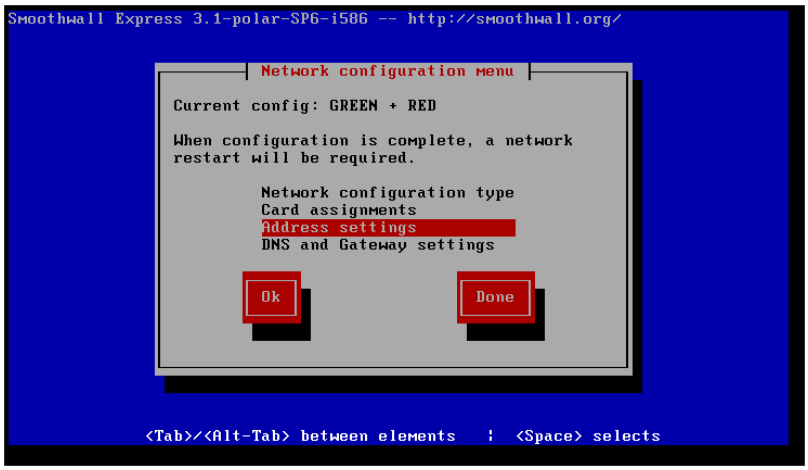


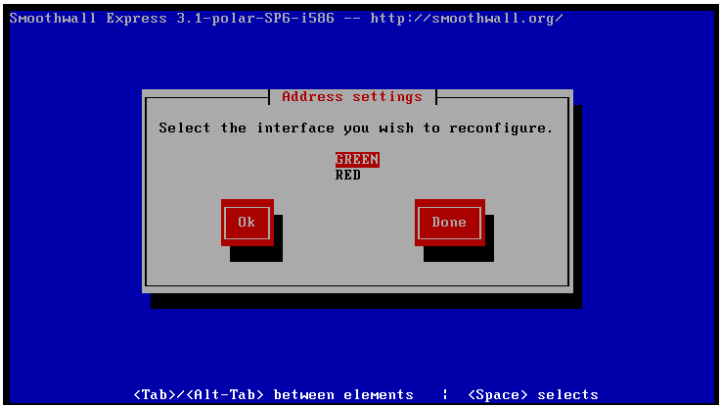
4. Confirm the network driver assignment:
    - GREEN: Intel(R) PRO/1000 Network Driver (eth0)
    - RED: Intel(R) PRO/1000 Network Driver (eth1)

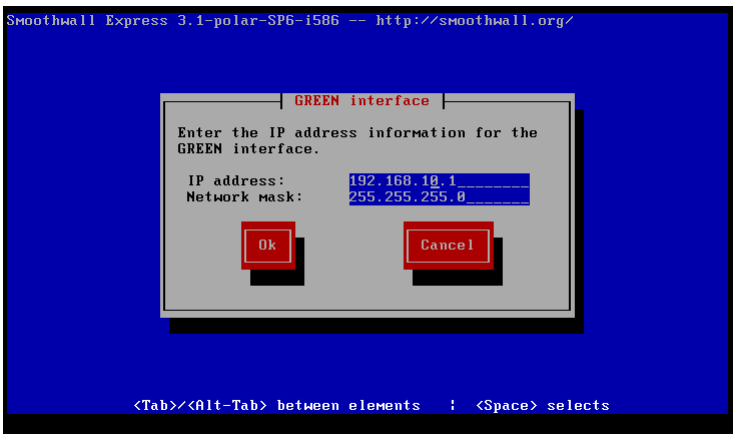




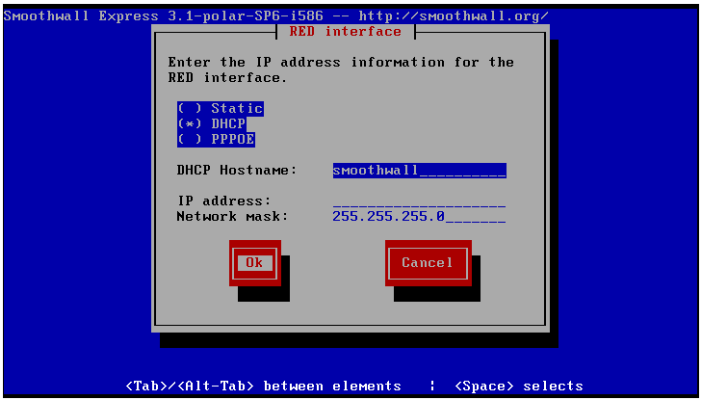
5. Go to **Address settings** → select **GREEN** interface → enter:
    - IP address: 192.168.10.1
    - Network mask: 255.255.255.0

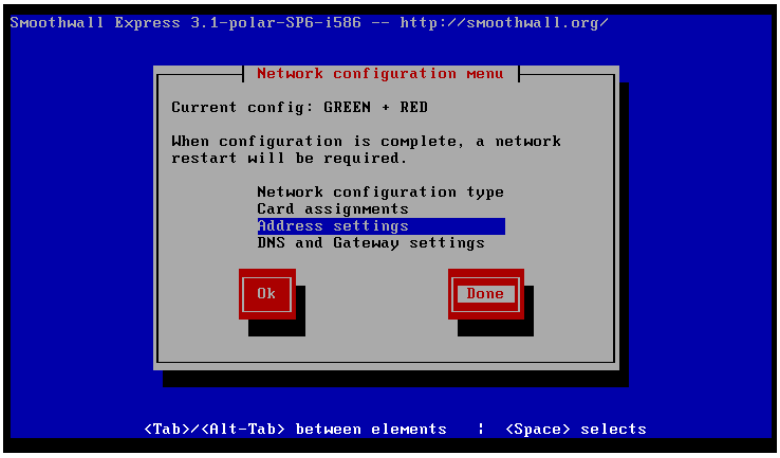




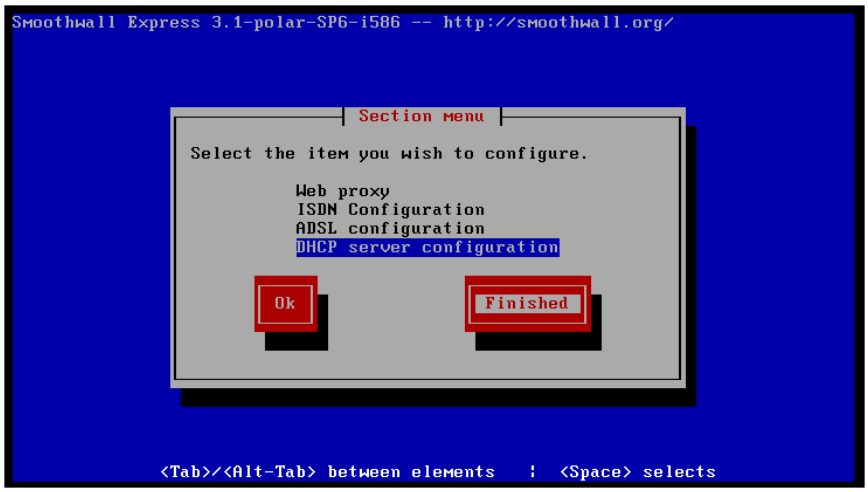


6. Select **RED** interface → choose connection type:
    - (*) DHCP
    - DHCP Hostname: smoothwall
    - Select Done

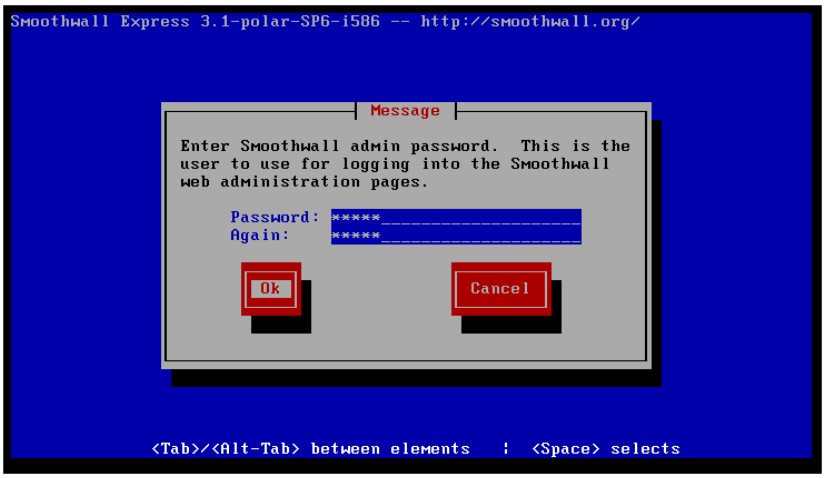




7. Return to main menu, go to **DHCP server configuration** (under Section Menu → Web proxy / ISDN / ADSL / DHCP server configuration), select **Finished**.



8. Set the Smoothwall **admin password** (used to log into the web administration pages) — enter and confirm password.

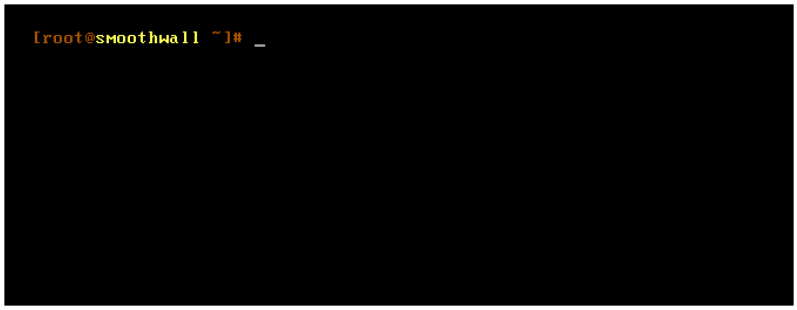


9. After setup completes, login prompt appears:

smoothwall login: root

Password:



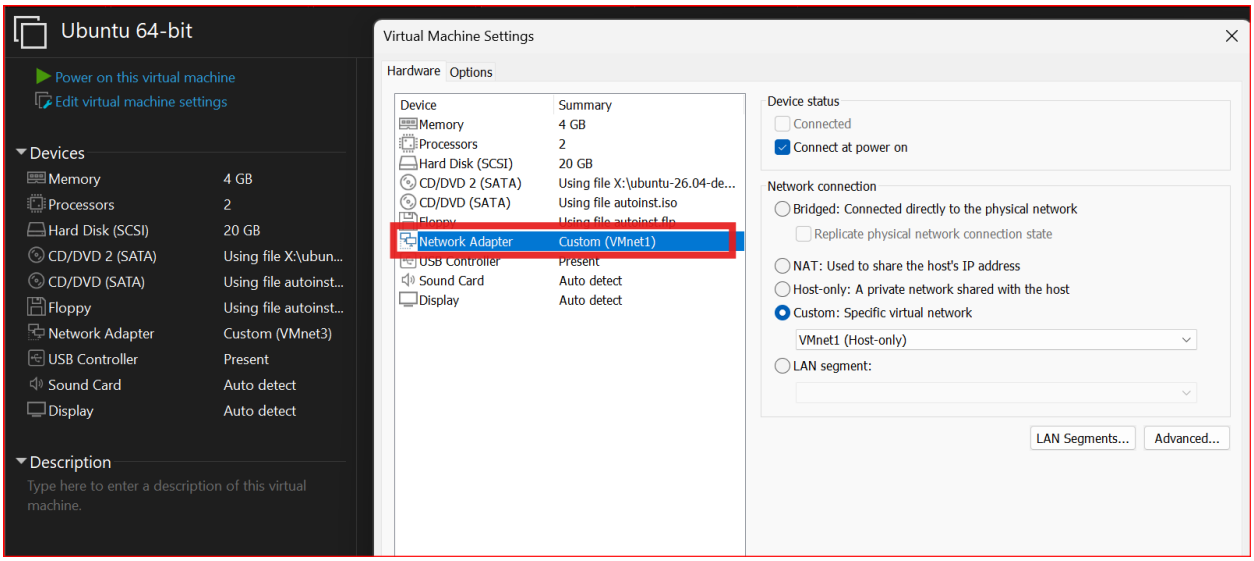


2.3 Verify Network with ip addr

```
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc sfq state UP group def
ault qlen 1000
    link/ether 00:0c:29:e1:15:b2 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::20c:29ff:fee1:15b2/64 scope link
       valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc sfq state UP group def
ault qlen 1000
    link/ether 00:0c:29:e1:15:bc brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.1/24 scope global eth1
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fee1:15bc/64 scope link
       valid_lft forever preferred_lft forever
4: ipsec0: <NOARP> mtu 0 qdisc noop state DOWN group default qlen 10
    link/void
5: ipsec1: <NOARP> mtu 0 qdisc noop state DOWN group default qlen 10
    link/void
6: mast0: <NOARP> mtu 0 qdisc noop state DOWN group default qlen 10
    link/none
7: teql0: <NOARP> mtu 1500 qdisc noop state DOWN group default qlen 100
    link/void
[root@smoothwall ~]#
```

3. Ubuntu "Operation Dept" VM — Network Setup

Open the Ubuntu VM used for the Operations Department. It should have its network adapter set to **Custom (VMnet1) — Host-only**.



3.1 Configure Static IP via Netplan

Open the netplan configuration file:

sudo nano /etc/netplan/01-netcfg.yaml

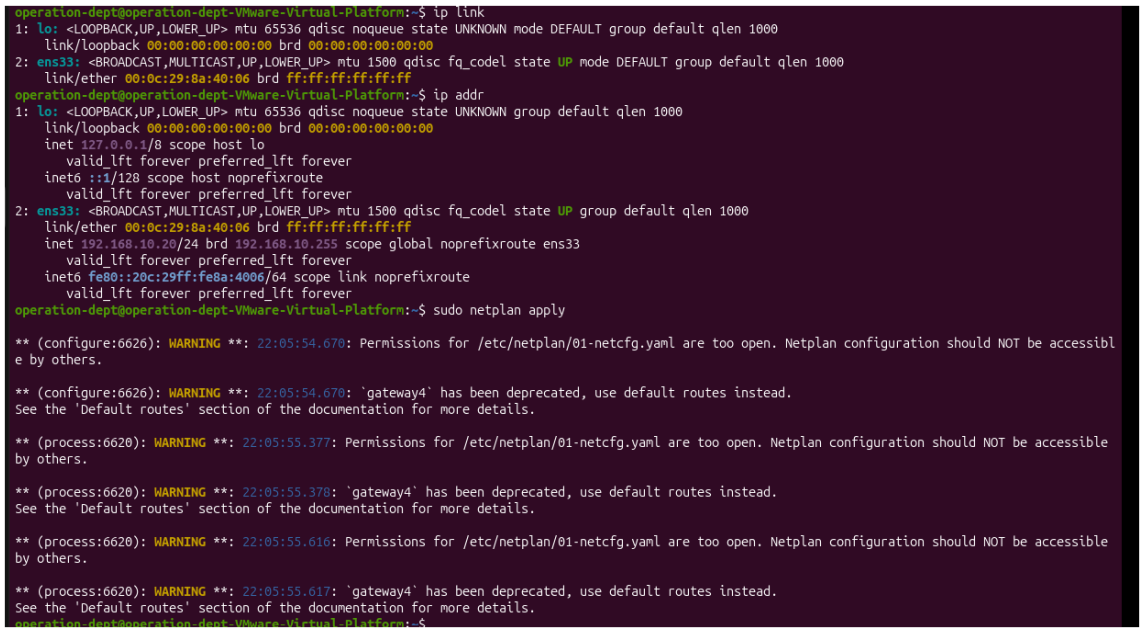
```
operation-dept@operation-dept-VMware-Virtual-Platform:~$ sudo nano /etc/netplan/
01-netcfg.yaml
```

```
network:
  version: 2
  ethernets:
    ens33:
      dhcp4: no
      addresses: [192.168.10.20/24]
      gateway4: 192.168.10.1
      nameservers:
        addresses: [192.168.10.10]
```

3.2 Apply and Verify Connectivity

Commands used:

- ip link
- ip addr
- sudo netplan apply

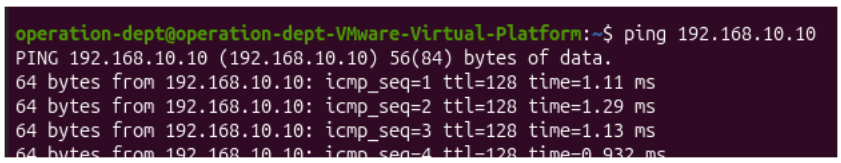


Result:

- Netplan warns that permissions on 01-netcfg.yaml are too open (should not be world-accessible) — tighten permissions if required.
- Netplan warns that gateway4 is deprecated in favor of default routes (still works, just a deprecation notice).
- Interface ens33 comes up with inet 192.168.10.20/24.

3.3 Check Connectivity to Domain Controller (DC01)

Power on DC01, then test connectivity:



Expected result: successful replies from 192.168.10.10 (ttl=128, low latency ~1ms).

LAB 1 — Linux Security Best Practices

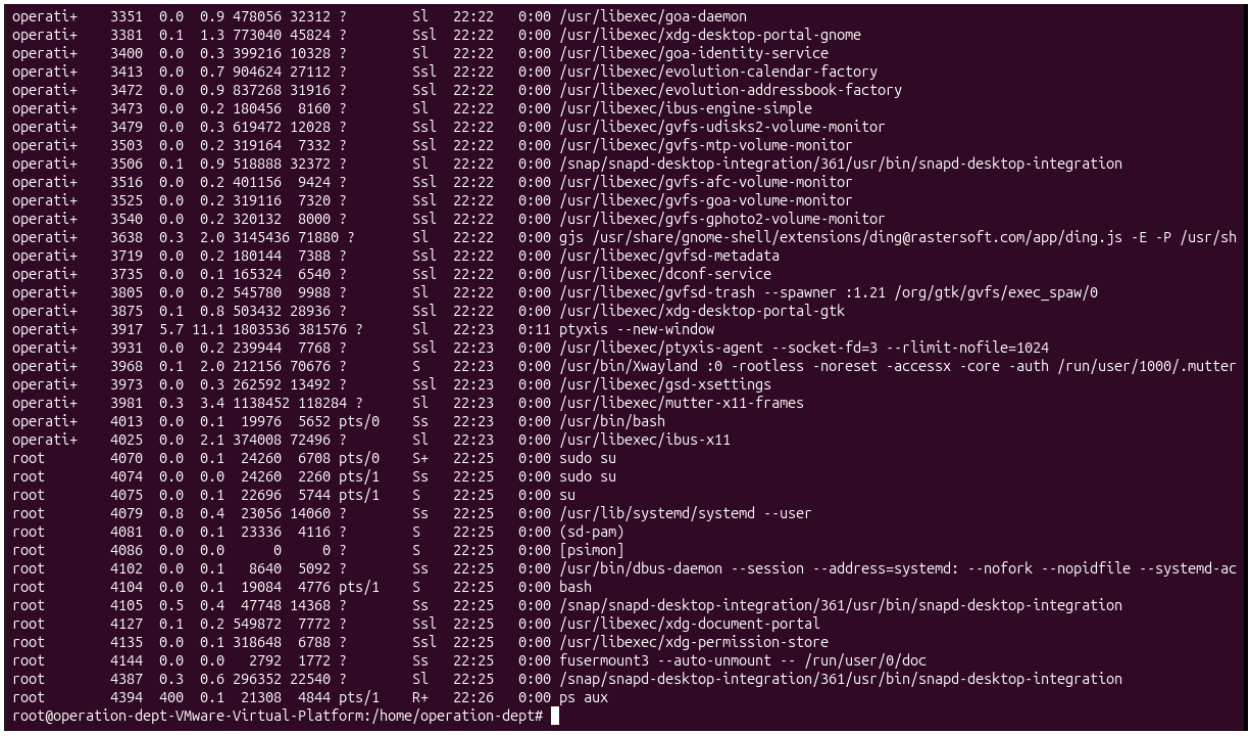
Executive Summary

Hardened the Ubuntu VM by disabling unnecessary services (Apache2, avahi-daemon) and cleaning unused packages. Implemented SSH hardening (disabled root login, empty passwords, and X11 forwarding; limited auth attempts and login grace time; restricted access to user "abdul"). Created a new user (abdul) with group membership, enforced password aging policies (90-day max, 7-day warning), tightened SSH file permissions, granted sudo access, generated SSH keys, and validated the hardened configuration via PuTTY login tests — confirming root login was successfully blocked.

1.1 List Running Services

ps aux

Review the list of running processes to identify unnecessary services.



```
operati+    3351  0.0  0.9 478056 32312 ?        Sl   22:22   0:00 /usr/libexec/goa-daemon
operati+    3381  0.1  1.3 773040 45824 ?        Ssl  22:22   0:00 /usr/libexec/xdg-desktop-portal-gnome
operati+    3400  0.0  0.3 399216 10328 ?        Sl   22:22   0:00 /usr/libexec/goa-identity-service
operati+    3413  0.0  0.7 904624 27112 ?        Ssl  22:22   0:00 /usr/libexec/evolution-calendar-factory
operati+    3472  0.0  0.9 837268 31916 ?        Ssl  22:22   0:00 /usr/libexec/evolution-addressbook-factory
operati+    3473  0.0  0.2 180456  8160 ?        Sl   22:22   0:00 /usr/libexec/ibus-engine-simple
operati+    3479  0.0  0.3 619472 12028 ?        Ssl  22:22   0:00 /usr/libexec/gvfs-udisks2-volume-monitor
operati+    3503  0.0  0.2 319164  7332 ?        Ssl  22:22   0:00 /usr/libexec/gvfs-mtp-volume-monitor
operati+    3506  0.1  0.9 518888 32372 ?        Sl   22:22   0:00 /snap/snapd-desktop-integration/361/usr/bin/snapd-desktop-integration
operati+    3516  0.0  0.2 401156  9424 ?        Ssl  22:22   0:00 /usr/libexec/gvfs-afc-volume-monitor
operati+    3525  0.0  0.2 319116  7320 ?        Ssl  22:22   0:00 /usr/libexec/gvfs-goa-volume-monitor
operati+    3540  0.0  0.2 320132  8000 ?        Ssl  22:22   0:00 /usr/libexec/gvfs-gphoto2-volume-monitor
operati+    3638  0.3  2.0 3145436 71880 ?       Sl   22:22   0:00 gjs /usr/share/gnome-shell/extensions/ding@rastersoft.com/app/ding.js -E -P /usr/sh
operati+    3719  0.0  0.2 180144  7388 ?        Ssl  22:22   0:00 /usr/libexec/gvfsd-metadata
operati+    3735  0.0  0.1 165324  6540 ?        Ssl  22:22   0:00 /usr/libexec/dconf-service
operati+    3805  0.0  0.2 545780  9988 ?        Sl   22:22   0:00 /usr/libexec/gvfsd-trash --spawner :1.21 /org/gtk/gvfs/exec_spaw/0
operati+    3875  0.1  0.8 503432 28936 ?        Ssl  22:22   0:00 /usr/libexec/xdg-desktop-portal-gtk
operati+    3917  5.7 11.1 1803536 381576 ?      Sl   22:23   0:11 ptyxis --new-window
operati+    3931  0.0  0.2 239944  7768 ?        Ssl  22:23   0:00 /usr/libexec/ptyxis-agent --socket-fd=3 --rlimit-nofile=1024
operati+    3968  0.1  2.0 212156 70676 ?        S    22:23   0:00 /usr/bin/Xwayland :0 -rootless -noreset -accessx -core -auth /run/user/1000/.mutter
operati+    3973  0.0  0.3 262592 13492 ?        Ssl  22:23   0:00 /usr/libexec/gsd-xsettings
operati+    3981  0.3  3.4 1138452 118284 ?      Sl   22:23   0:00 /usr/libexec/mutter-x11-frames
operati+    4013  0.0  0.1  19976  5652 pts/0    Ss   22:23   0:00 /usr/bin/bash
operati+    4025  0.0  2.1 374008 72496 ?        Sl   22:23   0:00 /usr/libexec/ibus-x11
root        4070  0.0  0.1  24260  6708 pts/0    S+   22:25   0:00 sudo su
root        4074  0.0  0.0  24260  2260 pts/1    Ss   22:25   0:00 sudo su
root        4075  0.0  0.1  22696  5744 pts/1    S    22:25   0:00 su
root        4079  0.8  0.4  23056 14060 ?        Ss   22:25   0:00 /usr/lib/systemd/systemd --user
root        4081  0.0  0.1  23336  4116 ?        S    22:25   0:00 (sd-pam)
root        4086  0.0  0.0      0     0 ?        S    22:25   0:00 [psimon]
root        4102  0.0  0.1   8640  5092 ?        Ss   22:25   0:00 /usr/bin/dbus-daemon --session --address=systemd: --nofork --nopidfile --systemd-ac
root        4104  0.0  0.1  19084  4776 pts/1    S    22:25   0:00 bash
root        4105  0.5  0.4  47748 14368 ?        Ss   22:25   0:00 /snap/snapd-desktop-integration/361/usr/bin/snapd-desktop-integration
root        4127  0.1  0.2 549872  7772 ?        Ssl  22:25   0:00 /usr/libexec/xdg-document-portal
root        4135  0.0  0.1 318648  6788 ?        Ssl  22:25   0:00 /usr/libexec/xdg-permission-store
root        4144  0.0  0.0   2792  1772 ?        Ss   22:25   0:00 fusermount3 --auto-unmount -- /run/user/0/doc
root        4387  0.3  0.6 296352 22540 ?        Sl   22:25   0:00 /snap/snapd-desktop-integration/361/usr/bin/snapd-desktop-integration
root        4394  400  0.1  21308  4844 pts/1    R+   22:26   0:00 ps aux
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept#
```

1.2 Disable Unnecessary Services

Disable Apache2 (if it exists):

- sudo systemctl disable apache2
- sudo systemctl stop apache2
- sudo update-rc.d -f apache2 remove

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo systemctl disable apache2
Failed to disable unit: Unit apache2.service does not exist
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo systemctl stop apache2
Failed to stop apache2.service: Unit apache2.service not loaded.
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo update-rc.d -f apache2 remove
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept#
```

Disable avahi-daemon:

- sudo apt remove avahi-daemon -y

This removes avahi-daemon, ipp-usb, and related dependent packages, and masks the avahi-daemon service.

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo apt remove avahi-daemon -y
The following packages were automatically installed and are no longer required:
  libavahi-core7          linux-headers-7.0.0-14-generic           linux-modules-7.0.0-14-generic
  libdaemon0              linux-image-unsigned-7.0.0-14-generic    linux-tools-7.0.0-14
  linux-headers-7.0.0-14  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14-generic
Use 'sudo apt autoremove' to remove them.

REMOVING:
  avahi-daemon  ipp-usb

Summary:
  Upgrading: 0, Installing: 0, Removing: 2, Not Upgrading: 1
  Freed space: 7,327 kB

(Reading database… 191742 files and directories currently installed.)
Removing ipp-usb (0.9.31-1)…
Removing avahi-daemon (0.8-18ubuntu1.1)…
Created symlink '/run/systemd/system/avahi-daemon.service' → '/dev/null'.
Masking 'avahi-daemon.service', but its triggering units are still active:
avahi-daemon.socket
Removed '/run/systemd/system/avahi-daemon.service'.
Processing triggers for dbus (1.16.2-2ubuntu4)…
Processing triggers for man-db (2.13.1-1build1)…
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo apt autoremove -y
REMOVING:
  libavahi-core7          linux-headers-7.0.0-14-generic           linux-modules-7.0.0-14-generic
  libdaemon0              linux-image-unsigned-7.0.0-14-generic    linux-tools-7.0.0-14
  linux-headers-7.0.0-14  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14-generic

Summary:
  Upgrading: 0, Installing: 0, Removing: 9, Not Upgrading: 1
  Freed space: 319 MB

(Reading database… 191695 files and directories currently installed.)
Removing libavahi-core7:amd64 (0.8-18ubuntu1.1)…
Removing libdaemon0:amd64 (0.14-7.1ubuntu5)…
Removing linux-headers-7.0.0-14-generic (7.0.0-14.14)…
Removing linux-headers-7.0.0-14 (7.0.0-14.14)…
Removing linux-tools-7.0.0-14-generic (7.0.0-14.14)…
Removing linux-tools-7.0.0-14 (7.0.0-14.14)…
```

Then clean unnecessary packages:

- sudo apt autoclean

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo apt autoclean
```

1.3 SSH Hardening — Restrict Root Login & Disable X11 Forwarding

Edit the SSH daemon config:

- sudo nano /etc/ssh/sshd_config

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo nano /etc/ssh/sshd_config
```

Set the following values:

PermitRootLogin no

PermitEmptyPasswords no

X11Forwarding no

```
  GNU nano 8.7.1
PermitRootLogin no
PermitEmptyPasswords no
X11Forwarding no
```

Restart SSH:

sudo systemctl restart ssh

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo nano /etc/ssh/sshd_config
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo systemctl restart ssh
```

1.4 Create a New User and Group

sudo addgroup developers

sudo adduser abdul

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo addgroup developers
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo adduser abdul
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
Sorry, passwords do not match.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
Sorry, passwords do not match.
New password:
BAD PASSWORD: The password fails the dictionary check - it is based on a dictionary word
Retype new password:
passwd: password updated successfully
Changing the user information for abdul
Enter the new value, or press ENTER for the default
        Full Name []: Abdul Wahid
        Room Number []: 699
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] y
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept#
```

Add the user to the group:

sudo usermod -aG developers abdul

groups abdul

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo usermod -aG developers abdul
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# groups abdul
abdul : abdul users developers
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept#
```

Expected output: abdul : abdul users developers

1.5 Password Aging & Expiry Policy

sudo chage -M 90 abdul

sudo chage -W 7 abdul

sudo chage -l abdul

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo chage -M 90 abdul
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo chage -W 7 abdul
```

Expected chage -l output includes:

Last password change : Jul 03, 2026

Password expires : Oct 01, 2026

Password inactive : never

Account expires : never

Minimum number of days between change : 0

Maximum number of days between change : 90

Number of days of warning before expiry: 7

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo chage -l abdul
Last password change                                    : Jul 03, 2026
Password expires                                        : Oct 01, 2026
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 90
Number of days of warning before password expires       : 7
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept#
```

1.6 SSH Access Policy Rules

Edit sshd_config again:

sudo nano /etc/ssh/sshd_config

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo nano /etc/ssh/sshd_config
```

Add/confirm these settings:

PermitRootLogin no

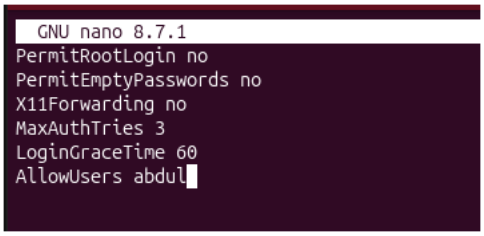
PermitEmptyPasswords no

X11Forwarding no

MaxAuthTries 3

LoginGraceTime 60

AllowUsers abdul



1.7 Check Config Syntax and Restart SSH

sudo sshd -t

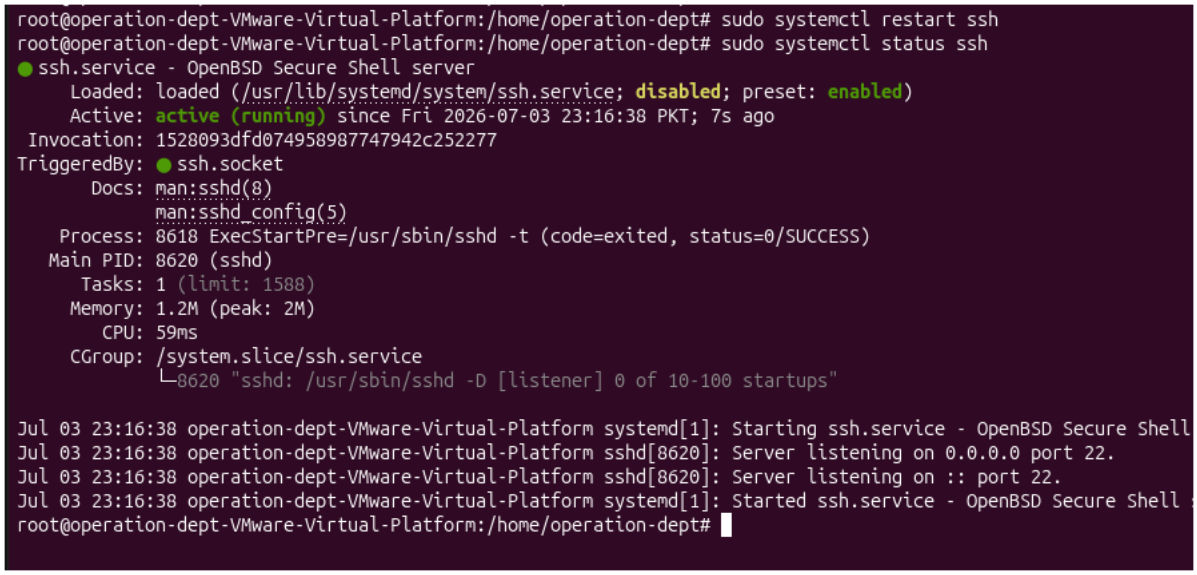
(should return no output if the config is valid)

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo sshd -t
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept#
```

Restart ssh and check its status

sudo systemctl restart ssh

sudo systemctl status ssh



1.8 UFW Firewall Setup

sudo apt-get install ufw -y

sudo ufw allow 22/tcp

sudo ufw default deny incoming

sudo ufw default allow outgoing

sudo ufw enable

sudo ufw status numbered

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo apt-get install ufw -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.2-9build1).
ufw set to manually installed.
Solving dependencies... Done
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo ufw allow 22/tcp
Rules updated
Rules updated (v6)
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo ufw enable
Firewall is active and enabled on system startup
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo ufw status numbered
Status: active

     To                         Action      From
     --                         ------      ----
[ 1] 22/tcp                     ALLOW IN    Anywhere
[ 2] 22/tcp (v6)                ALLOW IN    Anywhere (v6)

root@operation-dept-VMware-Virtual-Platform:/home/operation-dept#
```

Expected status:

Status: active

[1] 22/tcp ALLOW IN Anywhere

[2] 22/tcp (v6) ALLOW IN Anywhere (v6)

1.9 SSH Permissions Hardening

sudo chmod 600 /etc/ssh/sshd_config

sudo chmod 700 ~/.ssh

sudo chmod 600 ~/.ssh/authorized_keys

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo chmod 600 /etc/ssh/sshd_config
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo chmod 700 ~/.ssh
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo chmod 600 ~/.ssh/authorized_keys
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept#
```

Verify:

ls -la ~/.ssh/

stat /etc/ssh/sshd_config

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# ls -la ~/.ssh/
total 8
drwx------ 2 root root 4096 Jul  3 18:33 .
drwx------ 6 root root 4096 Jul  3 22:38 ..
-rw------- 1 root root    0 Jul  3 18:33 authorized_keys
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# stat /etc/ssh/sshd_config
  File: /etc/ssh/sshd_config
  size: 110             Blocks: 8          IO Block: 4096   regular file
Device: 8,2     Inode: 656100      Links: 1
Access: (0600/-rw-------)  Uid: (    0/    root)   Gid: (    0/    root)
Access: 2026-07-03 23:15:53.236216481 +0500
Modify: 2026-07-03 23:15:03.004681052 +0500
Change: 2026-07-03 23:20:56.499656714 +0500
 Birth: 2026-07-03 22:42:09.257152338 +0500
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept#
```

1.10 Sudo Access for Abdul

sudo visudo

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo visudo
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept#
```

Add this line under "User privilege specification":

abdul   ALL=(ALL:ALL) ALL

```
# Completely harmless preservation of a user preference.
#Defaults:%sudo env_keep += "GREP_COLOR"

# While you shouldn't normally run git as root, you need to with etckeep
#Defaults:%sudo env_keep += "GIT_AUTHOR_* GIT_COMMITTER_*"

# Per-user preferences; root won't have sensible values for them.
#Defaults:%sudo env_keep += "EMAIL DEBEMAIL DEBFULLNAME"

# "sudo scp" or "sudo rsync" should be able to use your SSH agent.
#Defaults:%sudo env_keep += "SSH_AGENT_PID SSH_AUTH_SOCK"

# Ditto for GPG agent
#Defaults:%sudo env_keep += "GPG_AGENT_INFO"

# Host alias specification

# User alias specification

# Cmnd alias specification

# User privilege specification
root    ALL=(ALL:ALL) ALL
abdul   ALL=(ALL:ALL) ALL
# Members of the admin group may gain root privileges
%admin ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:

@includedir /etc/sudoers.d
```

Verify:

sudo visudo -c

sudo grep "abdul" /etc/sudoers

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo visudo -c
/etc/sudoers: parsed OK
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo grep "abdul" /etc/sudoers
abdul   ALL=(ALL:ALL) ALL
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept#
```

Expected: /etc/sudoers: parsed OK and abdul ALL=(ALL:ALL) ALL

1.11 Create SSH Directory & Generate SSH Key for Abdul

sudo mkdir -p /home/abdul/.ssh

sudo chown abdul:abdul /home/abdul/.ssh

sudo chmod 700 /home/abdul/.ssh

ssh-keygen -t rsa -b 4096 -f /home/abdul/.ssh/id_rsa -N "abdul"

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo mkdir -p /home/abdul/.ssh
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo chown abdul:abdul /home/abdul/.ssh
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo chmod 700 /home/abdul/.ssh
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# ssh-keygen -t rsa -b 4096 -f /home/abdul/.ssh/id_rsa -N "abdul"
Generating public/private rsa key pair.
Your identification has been saved in /home/abdul/.ssh/id_rsa
Your public key has been saved in /home/abdul/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:uzAtC5LYSlOieMx8SWJfxzfslpeD1GKaacwak0lZUFI root@operation-dept-VMware-Virtual-Platform
The key's randomart image is:
+---[RSA 4096]----+
|      ooE        |
|       o         |
|        .        |
|       + . .     |
| .o.. + S B .    |
|oBo= + B X = .   |
|+oO = O X = +    |
|.o.o . X o . .   |
|.     o .        |
+----[SHA256]-----+
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept#
```

Set permissions:

sudo chmod 600 /home/abdul/.ssh/id_rsa

sudo chmod 600 /home/abdul/.ssh/id_rsa.pub

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo chmod 600 /home/abdul/.ssh/id_rsa
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo chmod 600 /home/abdul/.ssh/id_rsa.pub
```

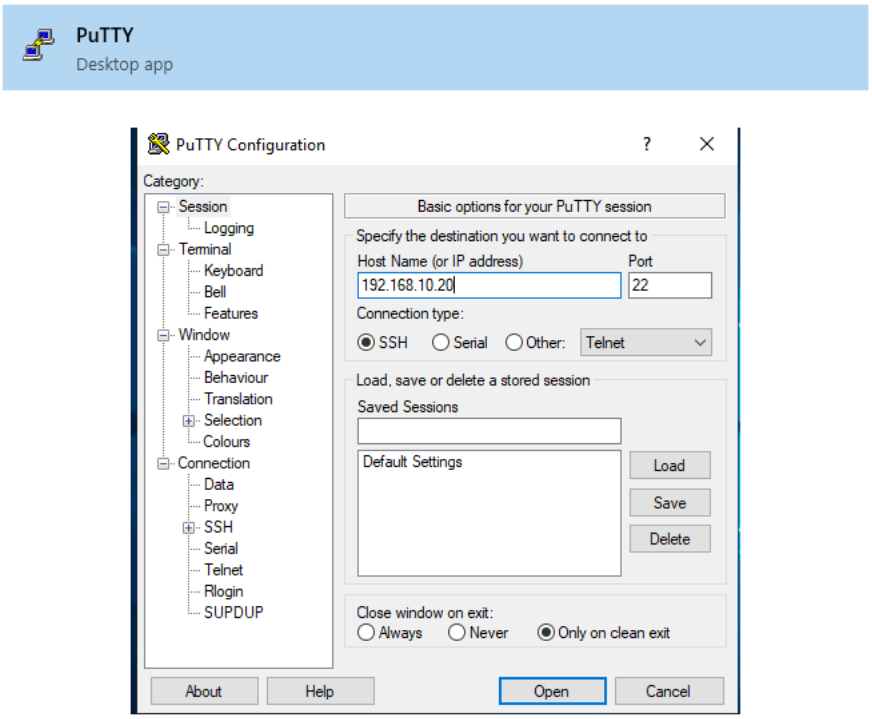
Verify:

sudo ls -la /home/abdul/.ssh/

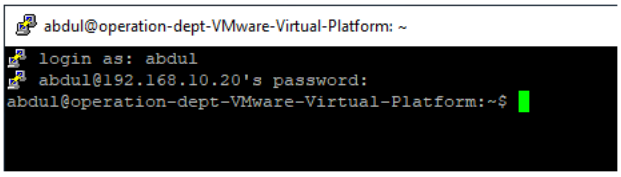
```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo ls -la /home/abdul/.ssh/
total 16
drwx------ 2 abdul abdul 4096 Jul  3 23:32 .
drwxr-x--- 3 abdul abdul 4096 Jul  3 23:31 ..
-rw------- 1 root  root  3454 Jul  3 23:32 id_rsa
-rw------- 1 root  root   769 Jul  3 23:32 id_rsa.pub
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept#
```

1.12 Test Login via PuTTY

Open PuTTY on DC01, connect to 192.168.10.20, port 22, SSH connection type.



Login as abdul with the created password.



Check user details:

whoami

groups

sudo ufw status numbered

sudo chage -l abdul

```
abdul@operation-dept-VMware-Virtual-Platform:~$ whoami
abdul
abdul@operation-dept-VMware-Virtual-Platform:~$ groups
abdul users developers
abdul@operation-dept-VMware-Virtual-Platform:~$ sudo ufw status numbered
[sudo: authenticate] Password:
Status: active

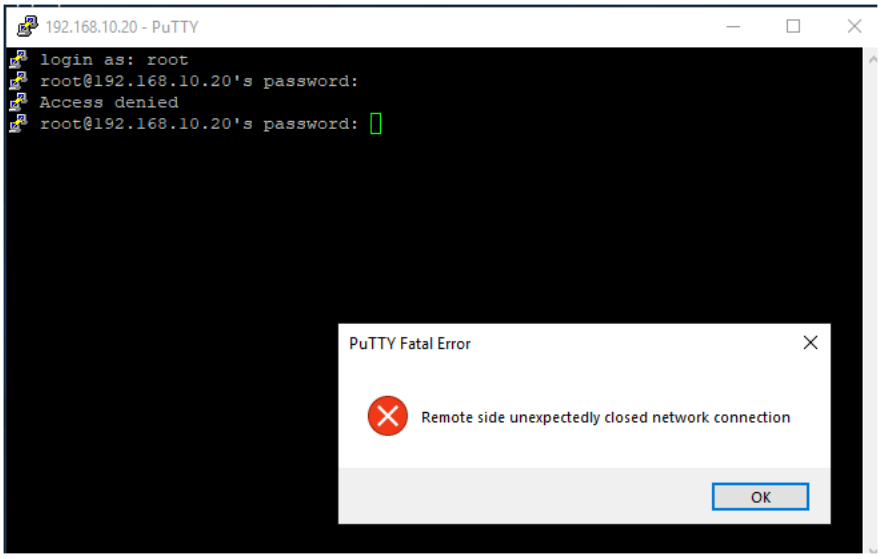
     To                         Action      From
     --                         ------      ----
[ 1] 22/tcp                     ALLOW IN    Anywhere
[ 2] 22/tcp (v6)                ALLOW IN    Anywhere (v6)

abdul@operation-dept-VMware-Virtual-Platform:~$ sudo chage -l abdul
Last password change                                    : Jul 03, 2026
Password expires                                        : Oct 01, 2026
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 90
Number of days of warning before password expires       : 7
abdul@operation-dept-VMware-Virtual-Platform:~$
```

Type exit to log out.

Now try logging in as root — access should be denied (root login disabled), confirming the SSH hardening worked:

login as: root root@192.168.10.20's password: Access denied



PuTTY will show a fatal error: "Remote side unexpectedly closed network connection."

LAB 2 — Implementation of Password Policies

Executive summary

Installed and configured libpam-pwquality to enforce strong password complexity (12-character minimum, mixed character requirements) and tested policy enforcement by rejecting a weak password. Configured /etc/login.defs for password aging, secure UMASK, and login logging, and set sudoers timeout and security defaults via visudo. Verified all policies were correctly applied across PAM, login.defs, and sudoers.

2.1 Install pwquality

sudo apt-get install libpam-pwquality -y

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo apt-get install libpam-pwquality -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
libpam-pwquality is already the newest version (1.4.5-5build1).
```

Edit the PAM config (/etc/pam.d/common-password) to include:

password requisite pam_pwquality.so retry=3 minlen=12 dcredit=-1 ucredit=-1 ocredit=-1 lcredit=-1

```
# here are the per-package modules (the "Primary" block)
password        requisite                       pam_pwquality.so retry=3 minlen=12  dcredit=-1 ucredit=-1 ocredit=-1 1credit=-1
password        [success=2 default=ignore]      pam_unix.so obscure use_authtok try_first_pass yescrypt
password        sufficient                      pam_sss.so use_authtok
# here's the fallback if no module succeeds
password        requisite                       pam_deny.so
# prime the stack with a positive return value if there isn't one already;
# this avoids us returning an error just because nothing sets a success code
# since the modules above will each just jump around
password        required                        pam_permit.so
# and here are more per-package modules (the "Additional" block)
password        optional        pam_gnome_keyring.so
# end of pam-auth-update config
```

2.2 Policy Check — Try a Weak Password

sudo passwd abdul

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo passwd abdul
New password:
BAD PASSWORD: The password contains less than 1 uppercase letters
Retype new password:
```

Example rejection: "BAD PASSWORD: The password contains less than 1 uppercase letters"

2.3 Login Defense Configuration (/etc/login.defs)

sudo nano /etc/login.defs

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo nano /etc/login.defs
```

Key settings:

PASS_MAX_DAYS 90

PASS_MIN_DAYS 1

PASS_WARN_AGE 7

UMASK 0077

LOG_UNKFAIL_ENAB yes

LOG_OK_LOGINS yes

UID_MIN 1000

UID_MAX 60000

```
#
# Password aging controls:
#
#       PASS_MAX_DAYS   Maximum number of days a password may be used.
#       PASS_MIN_DAYS   Minimum number of days allowed between password changes.
#       PASS_WARN_AGE   Number of days warning given before a password expires.
#
PASS_MAX_DAYS   90
PASS_MIN_DAYS   1
PASS_WARN_AGE   7
UMASK           0077
LOG_UNKFAIL_ENAB yes
LOG_OK_LOGINS    yes

#
# Min/max values for automatic uid selection in useradd(8)
#
UID_MIN                  1000
UID_MAX                 60000
# System accounts
```

2.4 Sudoers Timeout Settings

sudo visudo

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo visudo
```

Confirm/verify these Defaults lines exist:

Defaults env_reset

Defaults mail_badpass

Defaults secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"

Defaults use_pty

Defaults timestamp_timeout=5

```
Defaults        env_reset
Defaults        mail_badpass
Defaults        secure_path="/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/snap/bin"

# This fixes CVE-2005-4890 and possibly breaks some versions of kdesu
# (#1011624, https://bugs.kde.org/show_bug.cgi?id=452532)
Defaults        use_pty
Defaults        timestamp_timeout=5
# This preserves proxy settings from user environments of root
# equivalent users (group sudo)
```

2.5 Verify All Password Policies

sudo chage -l abdul

sudo grep "pam_pwquality" /etc/pam.d/common-password

sudo grep -E "PASS_MAX_DAYS|UMASK" /etc/login.defs

sudo grep "timestamp_timeout" /etc/sudoers

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo chage -l abdul
Last password change                                    : Jul 03, 2026
Password expires                                        : Oct 01, 2026
Password inactive                                       : never
Account expires                                         : never
Minimum number of days between password change          : 0
Maximum number of days between password change          : 90
Number of days of warning before password expires       : 7
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# grep "pam_pwquality" /etc/pam.d/common-password
password        requisite                       pam_pwquality.so retry=3 minlen=12  dcredit=-1 ucredit=-1 ocredit=-1 1credit=-1
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# grep -E "PASS_MAX_DAYS|UMASK" /etc/login.defs
#       PASS_MAX_DAYS   Maximum number of days a password may be used.
PASS_MAX_DAYS   90
UMASK           0077
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo grep "timestamp_timeout" /etc/sudoers
Defaults        timestamp_timeout=5
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept#
```

LAB 3 — Installing Lynis Security Auditing

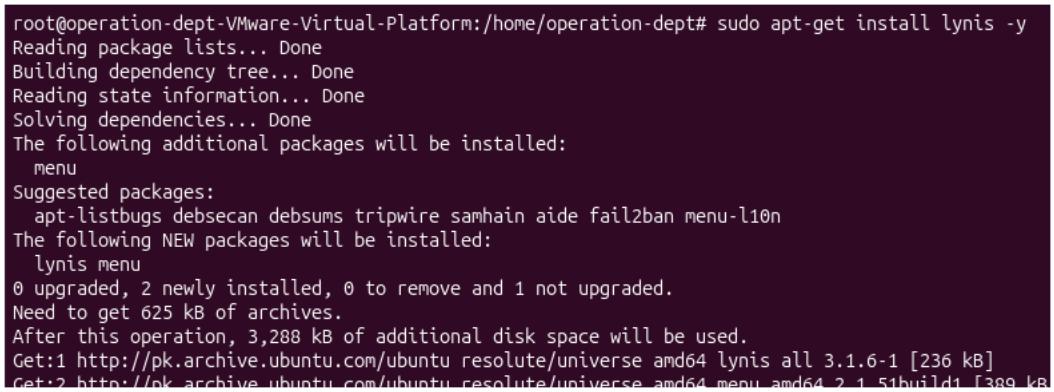
Executive summary

Installed Lynis (v3.1.6) and ran a full system security audit, generating a report (OS details, hostname, hardening index) with an initial score of 68/100. Installed a malware scanner (rkhunter) to address a Lynis-flagged gap, ran an initial rootkit/malware check (no rootkits or infections found), and re-ran Lynis to confirm the hardening score improved to 69/100.

3.1 Install and Check Version

sudo apt-get install lynis -y

lynis --version



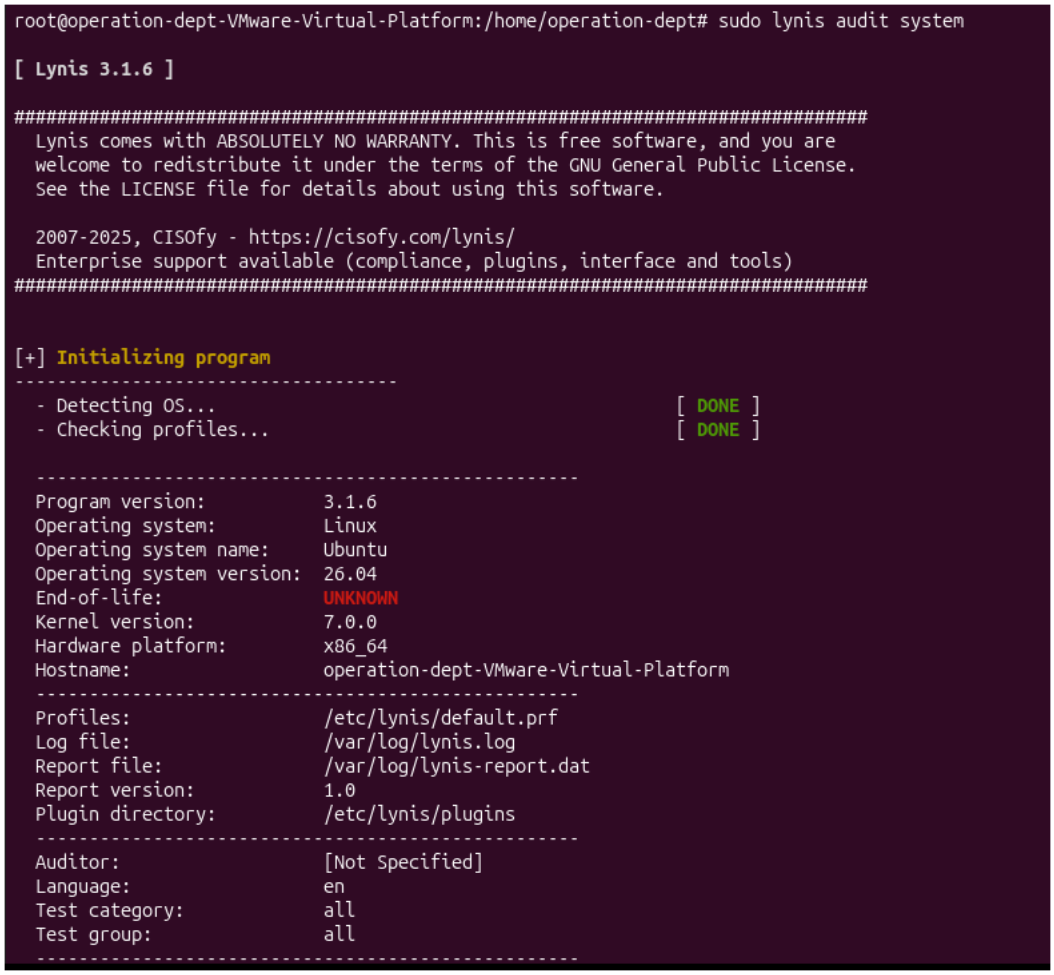


Output: Lynis 3.1.6

3.2 Run a Full System Audit

sudo lynis audit system

This scan takes 2–3 minutes and reports OS details, hostname, log file locations (/var/log/lynis.log, /var/log/lynis-report.dat), and runs all test categories.



3.3 View the Security Audit Report

sudo lynis show report

cat /var/log/lynis-report.dat

Report includes OS info, kernel version, hostname, VM type (vmware), and host IDs.

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo lynis show report
/var/log/lynis-report.dat
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# cat /var/log/lynis-report.dat
# Lynis Report
report_version_major=1
report_version_minor=0
report_datetime_start=2026-07-04 18:42:19
auditor=[Not Specified]
lynis_version=3.1.6
os=Linux
os_name=Ubuntu
os_fullname=Ubuntu 26.04 LTS
os_version=26.04
linux_version=Ubuntu
os_kernel_version=7.0.0
os_kernel_version_full=7.0.0-27-generic
hostname=operation-dept-VMware-Virtual-Platform
test_category=all
test_group=all
plugin_directory=/etc/lynis/plugins
lynis_update_available=0
vm=1
vmtype=vmware
container=0
notebook=0
systemd=1
hostid=19dfbaaed91f091c57094ec76a04c447b7fc98a9
hostid2=994624f224823c14fec14160210f6a5f9ce1ba6c907584345d3a0609a7f3835f
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept#
```

3.4 Cleaner Re-run and Log Review

sudo lynis audit system --quiet

sudo cat /var/log/lynis.log | tail -50

The quiet run only shows warnings, e.g. long execution time warnings.

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo lynis audit system --quiet
  [WARNING]: Test DEB-0001 had a long execution: 50.104404 seconds

  [WARNING]: Test PKGS-7345 had a long execution: 28.420720 seconds

  [WARNING]: Test PKGS-7392 had a long execution: 12.508372 seconds

  [WARNING]: Test CRYP-7902 had a long execution: 25.265370 seconds
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept#
```

Security logs best for searching for security scoring and findings

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo cat /var/log/lynis.log | tail -50
2026-07-04 18:47:07 Note: Using real readlink binary to determine symlink on /usr/bin/as
2026-07-04 18:47:07 Result: readlink shows /usr/bin/x86_64-linux-gnu-as as output
2026-07-04 18:47:07 Result: symlink found, pointing to file /usr/bin/x86_64-linux-gnu-as
2026-07-04 18:47:07 Binary: found /usr/bin/x86_64-linux-gnu-as (world executable)
2026-07-04 18:47:07 Hardening: assigned partial number of hardening points (2 of 3). Currently having 181 points (out of 261)
2026-07-04 18:47:07 Result: at least one compiler could be better hardened by restricting executable access to root or group only
2026-07-04 18:47:07 Suggestion: Harden compilers like restricting access to root user only [test:HRDN-7222] [details:-] [solution:-]
2026-07-04 18:47:07 ====
2026-07-04 18:47:07 Performing test ID HRDN-7230 (Check for malware scanner)
2026-07-04 18:47:07 Test: Check if a malware scanner is installed
2026-07-04 18:47:07 Result: no malware scanner found
2026-07-04 18:47:07 Suggestion: Harden the system by installing at least one malware scanner, to perform periodic file system scans [test:HRDN-7230] [
details:-] [solution:Install a tool like rkhunter, chkrootkit, OSSEC, Wazuh]
2026-07-04 18:47:07 Hardening: assigned partial number of hardening points (1 of 3). Currently having 182 points (out of 264)
2026-07-04 18:47:07 Result: no malware scanner found
2026-07-04 18:47:07 ====
2026-07-04 18:47:08 Performing test ID HRDN-7231 (Check for registered non-native binary formats)
2026-07-04 18:47:08 Test: Check for registered non-native binary formats
2026-07-04 18:47:08 Result: found 1 non-native binary formats registered: python3.14
2026-07-04 18:47:08 ====
2026-07-04 18:47:08 Action: Performing tests from category: Custom tests
2026-07-04 18:47:08 Test: Checking for tests_custom file
2026-07-04 18:47:08 ====
2026-07-04 18:47:08 Action: Performing plugin tests
2026-07-04 18:47:08 Result: Found 1 plugins of which 1 are enabled
2026-07-04 18:47:08 Result: Plugins phase 2 finished
2026-07-04 18:47:08 Checking permissions of /usr/share/lynis/include/report
2026-07-04 18:47:08 File permissions are OK
2026-07-04 18:47:08 Hardening index : [68] [#############       ]
2026-07-04 18:47:08 Hardening strength: System has been hardened, but could use additional hardening
2026-07-04 18:47:08 Checking permissions of /usr/share/lynis/include/tool_tips
2026-07-04 18:47:08 File permissions are OK
2026-07-04 18:47:08 Tool tips: enabled
2026-07-04 18:47:08 ================================================================================
2026-07-04 18:47:08 Tests performed:     250
2026-07-04 18:47:08 Total tests:         466
2026-07-04 18:47:08 Active plugins:      1
2026-07-04 18:47:08 Total plugins:       1
2026-07-04 18:47:08 ================================================================================
```

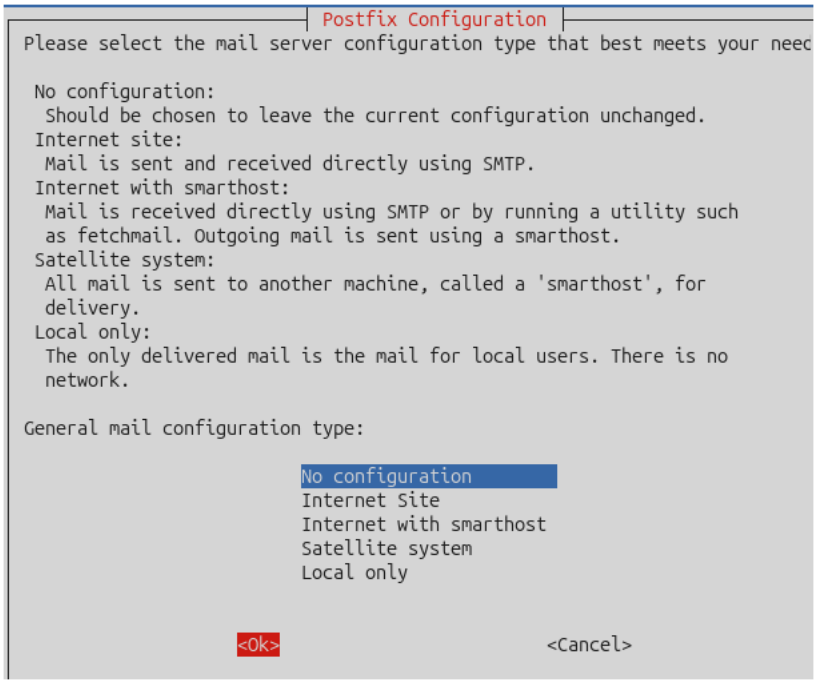
3.5 Install a Malware Scanner (rkhunter)

sudo apt-get install rkhunter -y

During install, a **Postfix Configuration** screen may appear — select **"No configuration"** and press OK.

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept# sudo apt-get install rkhunter -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following additional packages will be installed:
  bsd-mailx fonts-lato javascript-common libjs-jquery liblockfile-bin liblockfile1 libnsl2 libruby libruby3
  ruby-did-you-mean ruby-net-telnet ruby-ruby2-keywords ruby-rubygems ruby-sdbm ruby-webrick ruby-xmlrpc ru
  unhide.rb
Suggested packages:
  apache2 | lighttpd | httpd postfix-cdb postfix-doc postfix-ldap postfix-lmdb postfix-mta-sts-resolver pos
  postfix-pgsql postfix-sqlite procmail sasl2-bin | dovecot-common ri ruby-dev bundler
The following NEW packages will be installed:
  bsd-mailx fonts-lato javascript-common libjs-jquery liblockfile-bin liblockfile1 libnsl2 libruby libruby3
  ruby-csv ruby-did-you-mean ruby-net-telnet ruby-ruby2-keywords ruby-rubygems ruby-sdbm ruby-webrick ruby-
  unhide unhide.rb
0 upgraded, 26 newly installed, 0 to remove and 1 not upgraded.
Need to get 11.4 MB of archives.
After this operation, 52.1 MB of additional disk space will be used.
Get:1 http://pk.archive.ubuntu.com/ubuntu resolute/main amd64 fonts-lato all 2.015-1build1 [2,779 kB]
```

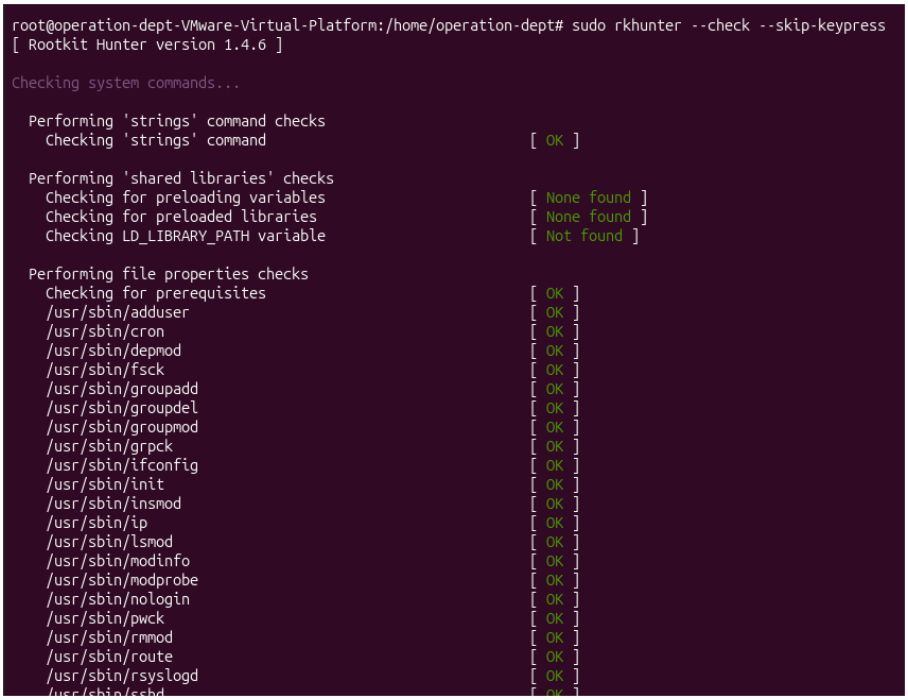
Click no configuration and select ok



3.6 Run Initial rkhunter Scan

sudo rkhunter --check --skip-keypress

This performs command checks, shared library checks, file property checks, group/account checks, system configuration checks, and filesystem checks.

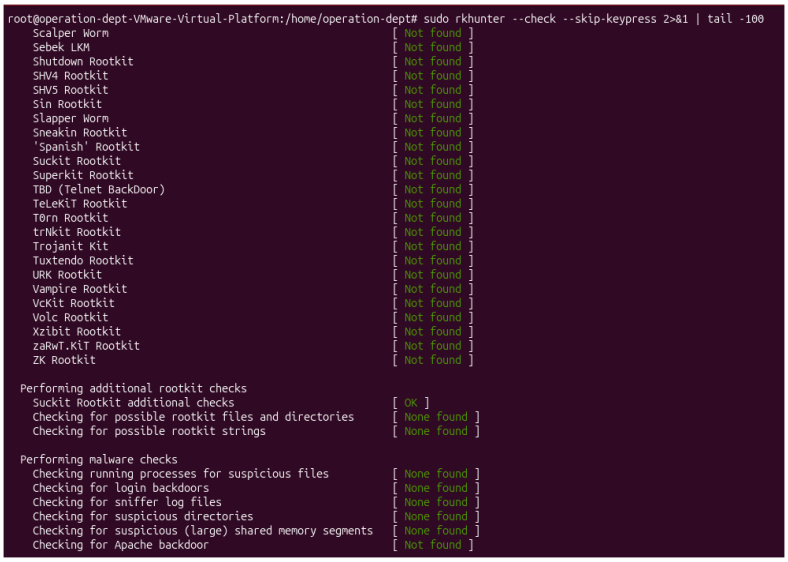


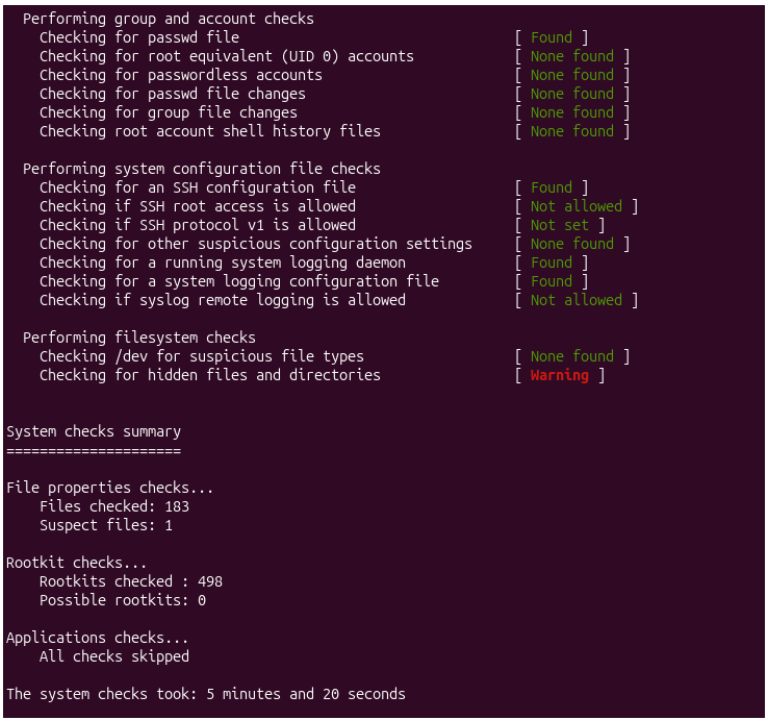




3.7 View Final Rootkit/Malware Report

sudo rkhunter --check --skip-keypress 2>&1 | tail -100





Final summary includes:

System checks summary

==========

File properties checks...

Files checked: 183

Suspect files: 1

Rootkit checks...

Rootkits checked : 498

Possible rootkits: 0

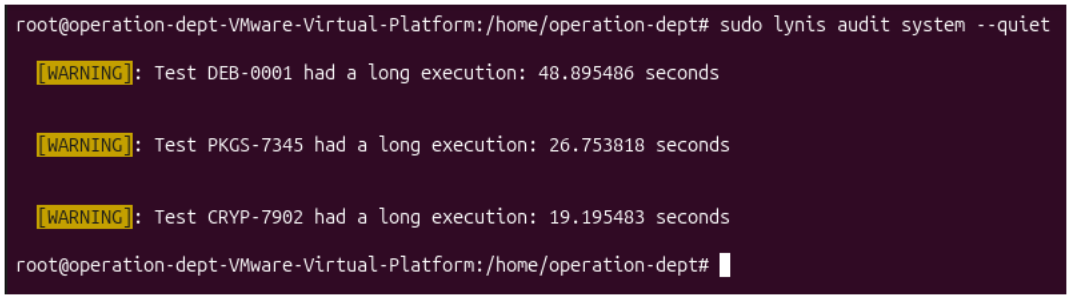
Applications checks...

All checks skipped

The system checks took: 5 minutes and 20 seconds

3.8 Re-run Lynis to Confirm Improved Hardening Score

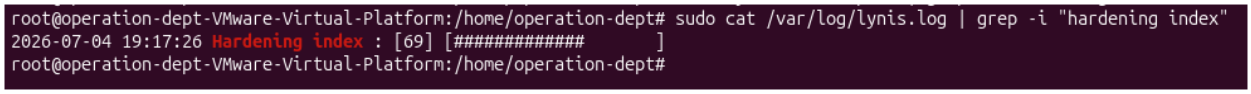
sudo lynis audit system --quiet



sudo cat /var/log/lynis.log | grep -i "hardening index"

(This confirms the hardening score improved after installing the malware scanner and applying earlier fixes.)

Result:



LAB 4 — File Integrity Checking with Checksums (SHA-256)

Executive summary

Created test files and generated SHA-256 checksums to establish file integrity baselines. Demonstrated integrity verification (all files "OK") and successfully detected tampering after modifying a file (checksum mismatch flagged as "FAILED"). Extended this to checksum critical system files (/etc/passwd, /etc/shadow, /etc/sudoers, sshd_config) and built an automated integrity-check script, scheduled via cron to run daily at 2 AM, logging results for ongoing monitoring.

4.1 Create Test Files

echo "Secure file" > /tmp/secure.txt

echo "IMP file" > /tmp/imp.txt

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# echo "Secure file" > /tmp/secure.txt
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# echo "IMP file" > /tmp/imp.txt
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

4.2 Locate sha256sum

which sha256sum

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# which sha256sum
/usr/bin/sha256sum
```

Output: /usr/bin/sha256sum

4.3 Generate Checksums

/usr/bin/sha256sum /tmp/secure.txt > /tmp/checksums.sha256

/usr/bin/sha256sum /tmp/imp.txt >> /tmp/checksums.sha256

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# /usr/bin/sha256sum /tmp/secure.txt > /tmp/checksums.sha256
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# /usr/bin/sha256sum /tmp/imp.txt >> /tmp/checksums.sha256
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

4.4 Verify Checksum File Content

cat /tmp/checksums.sha256

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# cat /tmp/checksums.sha256
fda5ef59ccbb1fc632cc92a55c09a9c9b458fc847def3f5fe16e3a5e5a46fdd3  /tmp/secure.txt
fce8795d1652effa11201d0a211dc290e95b9accf92441af86e9040040c4c585  /tmp/imp.txt
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

Example output:

fda5ef59ccbb1fc632c92a55c09a9c9b458fc847def3f5fe16e3a5e5a46fdd3 /tmp/secure.txt

fce8795d1652effa11201d0a211dc290e95b9accf92441af86e9040040c4c585 /tmp/imp.txt

4.5 Check File Integrity

/usr/bin/sha256sum -c /tmp/checksums.sha256

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# /usr/bin/sha256sum -c /tmp/checksums.sha256
/tmp/secure.txt: OK
/tmp/imp.txt: OK
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

Both files should return OK.

4.6 Demonstrate Integrity Failure After Modification

echo "Modified the important file" >> /tmp/imp.txt

/usr/bin/sha256sum -c /tmp/checksums.sha256

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# echo "Modified the important  file" >> /tmp/imp.txt
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# /usr/bin/sha256sum -c /tmp/checksums.sha256
/tmp/secure.txt: OK
/tmp/imp.txt: FAILED
sha256sum: WARNING: 1 computed checksum did NOT match
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

Result:

/tmp/secure.txt: OK

/tmp/imp.txt: FAILED

sha256sum: WARNING: 1 computed checksum did NOT match

4.7 Create Two More Files

echo "This is a secure file" > /tmp/secure_file.txt

echo "Another important file" > /tmp/important_data.txt

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# echo "This is a secure file" > /tmp/secure_file.txt
echo "Another important file" > /tmp/important_data.txt
```

4.8 Checksum Critical System Files

sudo mkdir -p /etc/checksums

sudo /usr/bin/sha256sum /etc/passwd > /etc/checksums/system.sha256

sudo /usr/bin/sha256sum /etc/shadow >> /etc/checksums/system.sha256

sudo /usr/bin/sha256sum /etc/sudoers >> /etc/checksums/system.sha256

sudo /usr/bin/sha256sum /etc/ssh/sshd_config >> /etc/checksums/system.sha256

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo mkdir -p /etc/checksums

sudo /usr/bin/sha256sum /etc/passwd > /etc/checksums/system.sha256
sudo /usr/bin/sha256sum /etc/shadow >> /etc/checksums/system.sha256
sudo /usr/bin/sha256sum /etc/sudoers >> /etc/checksums/system.sha256
sudo /usr/bin/sha256sum /etc/ssh/sshd_config >> /etc/checksums/system.sha256
```

Verify:

sudo cat /etc/checksums/system.sha256

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo cat /etc/checksums/system.sha256
f20c372053cc1b2b18a568c7ce0ef15caa0406c753f8b0cfcefef0ff2d599a83  /etc/passwd
cfde0fd87aa9eb605a04ee93bcb24c2d2b7310a198c046f3da3ea7ca1fa33dd1  /etc/shadow
2ec3be82b97d6949870cc3fc27e1b73586bbcca5a6215505f35d47b567d57386  /etc/sudoers
787b22cf4b442ff1fff4ac4f0a3ffcff6c1b6eaa8b32df3432753a7ffb514d2d  /etc/ssh/sshd_config
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

Integrity check:

sudo /usr/bin/sha256sum -c /etc/checksums/system.sha256

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo /usr/bin/sha256sum -c /etc/checksums/system.sha256
/etc/passwd: OK
/etc/shadow: OK
/etc/sudoers: OK
/etc/ssh/sshd_config: OK
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

All four files should return OK.

4.9 Create an Automated Integrity Check Script

sudo nano /usr/local/bin/check_integrity.sh

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo nano /usr/local/bin/check_integrity.sh
```

Script content:

bash

#!/bin/bash

echo "===== File Integrity Check Report ====="

echo "Date: $(date)"

echo ""

Check system files

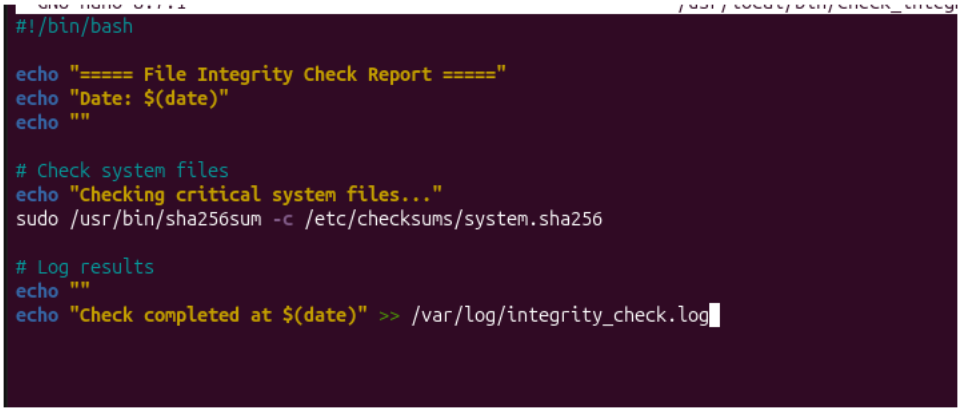
echo "Checking critical system files..."

sudo /usr/bin/sha256sum -c /etc/checksums/system.sha256

Log results

echo ""

echo "Check completed at $(date)" >> /var/log/integrity_che



4.10 Make Executable and Test

sudo chmod +x /usr/local/bin/check_integrity.sh

sudo /usr/local/bin/check_integrity.sh

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo chmod +x /usr/local/bin/check_integrity.sh
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo /usr/local/bin/check_integrity.sh
===== File Integrity Check Report =====
Date: Sat Jul  4 07:51:00 PM PKT 2026

Checking critical system files...
/etc/passwd: OK
/etc/shadow: OK
/etc/sudoers: OK
/etc/ssh/sshd_config: OK

root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

4.11 Schedule Daily Automated Check via Cron

sudo crontab -e

Choose editor option 1 (nano) if prompted.

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo crontab -e
no crontab for root - using an empty one
Select an editor.  To change later, run select-editor again.
  1. /bin/nano        <---- easiest
  2. /usr/bin/vim.tiny
  3. /bin/ed

Choose 1-3 [1]: 1
crontab: installing new crontab
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

Add this line at the end of the crontab file:

0 2 * * * /usr/local/bin/check_integrity.sh >> /var/log/integrity_check.log 2>&1

```
 GNU nano 8.7.1                                  /tmp/crontab.xDmKi5/crontab *
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 2 * * * /usr/local/bin/check_integrity.sh >> /var/log/integrity_check.log 2>&1
```

Verify:

sudo crontab -l

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 2 * * * /usr/local/bin/check_integrity.sh >> /var/log/integrity_check.log 2>&1
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

LAB 5 — Malware Scanning with ClamAV

Executive summary

Installed ClamAV, ClamAV-daemon, and ClamTK; resolved log/lock file permission issues and updated the virus database via freshclam. Scanned test directories (including an EICAR test file) and the user's home directory — no infections found. Built and scheduled an automated weekly malware scan script (Sundays at 3 AM) covering home and /tmp directories, with results logged and manually verified.

5.1 Install ClamAV and ClamTK

sudo apt-get install clamav clamav-daemon clamtk -y

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo apt-get install clamav clamav-daemon clamtk -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following additional packages will be installed:
  clamav-base clamav-freshclam clamdscan libclamav12 libcommon-sense-perl libjson-perl libjson-xs-perl libmspack0t64 libtext-csv-p
  libtext-csv-xs-perl libtypes-serialiser-perl
Suggested packages:
  libclamunrar clamav-doc daemon cabextract clamtk-gnome libclamunrar11
The following NEW packages will be installed:
  clamav clamav-base clamav-daemon clamav-freshclam clamdscan clamtk libclamav12 libcommon-sense-perl libjson-perl libjson-xs-perl
  libtext-csv-perl libtext-csv-xs-perl libtypes-serialiser-perl
0 upgraded, 14 newly installed, 0 to remove and 1 not upgraded.
```

5.2 Start and Enable ClamAV Daemon

sudo systemctl start clamav-daemon

sudo systemctl enable clamav-daemon

sudo systemctl status clamav-daemon

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo systemctl start clamav-daemon
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo systemctl enable clamav-daemon
Synchronizing state of clamav-daemon.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable clamav-daemon
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo systemctl status clamav-daemon
○ clamav-daemon.service - Clam AntiVirus userspace daemon
     Loaded: loaded (/usr/lib/systemd/system/clamav-daemon.service; enabled; preset: enabled)
    Drop-In: /etc/systemd/system/clamav-daemon.service.d
             └─extend.conf
     Active: inactive (dead)
TriggeredBy: ○ clamav-daemon.socket
  Condition: start condition unmet at Sat 2026-07-04 20:00:29 PKT; 21s ago
       Docs: man:clamd(8)
             man:clamd.conf(5)
             https://docs.clamav.net/

Jul 04 19:59:20 operation-dept-VMware-Virtual-Platform systemd[1]: clamav-daemon.service - Clam AntiVirus userspace daemon skipped
Jul 04 20:00:29 operation-dept-VMware-Virtual-Platform systemd[1]: clamav-daemon.service - Clam AntiVirus userspace daemon skipped
lines 1-13/13 (END)
```

5.3 Stop Clam, Fix Lock File and Log Directory Permissions

sudo systemctl stop clamav-daemon

sudo rm /var/log/clamav/freshclam.log

sudo mkdir -p /var/log/clamav

sudo chown clamav:clamav /var/log/clamav

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo systemctl stop clamav-daemon
```

Remove lock file

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo rm /var/log/clamav/freshclam.log
```

Create directory with proper permissions

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo mkdir -p /var/log/clamav
```

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo chown clamav:clamav /var/log/clamav
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

5.4 Update the Virus Database

sudo freshclam

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo freshclam
Sat Jul  4 20:03:53 2026 -> ClamAV update process started at Sat Jul  4 20:03:53 2026
Sat Jul  4 20:03:55 2026 -> daily.cvd database is up-to-date (version: 28050, sigs: 355485, f-level: 90, builder: svc.clamav-publis
Sat Jul  4 20:03:55 2026 -> main.cvd database is up-to-date (version: 63, sigs: 3287027, f-level: 90, builder: tomjudge)
Sat Jul  4 20:03:55 2026 -> bytecode.cvd database is up-to-date (version: 339, sigs: 80, f-level: 90, builder: nrandolp)
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

Output confirms daily.cvd, main.cvd, and bytecode.cvd databases are up to date.

5.5 Restart ClamAV Daemon

sudo systemctl start clamav-daemon

sudo systemctl status clamav-daemon

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo systemctl start clamav-daemon
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo systemctl status clamav-daemon
● clamav-daemon.service - Clam AntiVirus userspace daemon
     Loaded: loaded (/usr/lib/systemd/system/clamav-daemon.service; enabled; preset: enabled)
    Drop-In: /etc/systemd/system/clamav-daemon.service.d
             └─extend.conf
     Active: active (running) since Sat 2026-07-04 20:06:10 PKT; 24s ago
 Invocation: 046e7740545540618b040c55fa53bf70
TriggeredBy: ● clamav-daemon.socket
       Docs: man:clamd(8)
             man:clamd.conf(5)
             https://docs.clamav.net/
   Main PID: 479717 (clamd)
      Tasks: 1 (limit: 1588)
     Memory: 986.3M (peak: 986.5M, swap: 232K, swap peak: 232K)
        CPU: 18.887s
     CGroup: /system.slice/clamav-daemon.service
             └─479717 /usr/sbin/clamd --foreground=true

Jul 04 20:06:10 operation-dept-VMware-Virtual-Platform systemd[1]: Starting clamav-daemon.service - Clam AntiVirus userspace daemon
Jul 04 20:06:10 operation-dept-VMware-Virtual-Platform systemd[1]: Started clamav-daemon.service - Clam AntiVirus userspace daemon.
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

Expected: Active: active (running)

5.6 Create Test Files and Download EICAR Test File

mkdir -p /tmp/scan_test

echo "This is a safe file" > /tmp/scan_test/file1.txt

echo "Another safe file" > /tmp/scan_test/file2.txt

wget -O /tmp/scan_test/eicar.com http://www.eicar.org/download/eicar.com 2>/dev/null || echo "Test file"

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# mkdir -p /tmp/scan_test
```

Make test files

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# echo "This is a safe file" > /tmp/scan_test/file1.txt
echo "Another safe file" > /tmp/scan_test/file2.txt
```

Download ECAR Test file

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# wget -O /tmp/scan_test/eicar.com http://www.eicar.org/download/eicar.com 2>/
dev/null || echo "Test file"
Test file
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

Verify files:

ls -la /tmp/scan_test/

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# ls -la /tmp/scan_test/
total 8
drwx------  2 root root 100 Jul  4 20:08 .
drwxrwxrwt 17 root root 500 Jul  4 20:08 ..
-rw-------  1 root root   0 Jul  4 20:08 eicar.com
-rw-------  1 root root  20 Jul  4 20:08 file1.txt
-rw-------  1 root root  18 Jul  4 20:08 file2.txt
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

5.7 Scan the Test Directory

sudo clamscan -r /tmp/scan_test/

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo clamscan -r /tmp/scan_test/
Loading:    21s, ETA:   0s [=========================>]    3.63M/3.63M sigs
Compiling:   6s, ETA:   0s [=========================>]       41/41 tasks

/tmp/scan_test/eicar.com: Empty file
/tmp/scan_test/file2.txt: OK
/tmp/scan_test/file1.txt: OK

----------- SCAN SUMMARY -----------
Known viruses: 3627891
Engine version: 1.4.4
Scanned directories: 1
Scanned files: 2
Infected files: 0
Data scanned: 0.00 MB
Data read: 0.00 MB (ratio 0.00:1)
Time: 27.567 sec (0 m 27 s)
Start Date: 2026:07:04 20:10:57
End Date:   2026:07:04 20:11:24
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

5.8 Scan the User's Home Directory

sudo clamscan -r -i --remove /home/abdul/

(-i shows only infected files, --remove deletes infected files if found)

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo clamscan -r -i --remove /home/abdul/

----------- SCAN SUMMARY -----------
Known viruses: 3627891
Engine version: 1.4.4
Scanned directories: 2
Scanned files: 6
Infected files: 0
Data scanned: 0.00 MB
Data read: 0.00 MB (ratio 0.00:1)
Time: 18.761 sec (0 m 18 s)
Start Date: 2026:07:04 20:12:23
End Date:   2026:07:04 20:12:42
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

5.9 Create an Automated Weekly Malware Scan Script

sudo nano /usr/local/bin/weekly_malware_scan.sh

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo nano /usr/local/bin/weekly_malware_scan.sh
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

Script content:

bash

#!/bin/bash

echo "===== Weekly Malware Scan Report ====="

echo "Date: $(date)"

echo ""

Scan home directory

sudo clamscan -r /home/abdul/ >> /var/log/malware_scan.log **2>&1**

Scan /tmp

sudo clamscan -r /tmp/ >> /var/log/malware_scan.log **2>&1**

Log completion

echo "Scan completed at $(date)" >> /var/log/malware_scan.

```
 GNU nano 8.7.1                                  /usr/local/bin/weekly_malware_scan.sh *
#!/bin/bash

echo "===== Weekly Malware Scan Report ====="
echo "Date: $(date)"
echo ""

# Scan home directory
sudo clamscan -r /home/abdul/ >> /var/log/malware_scan.log 2>&1

# Scan /tmp
sudo clamscan -r /tmp/ >> /var/log/malware_scan.log 2>&1

# Log completion
echo "Scan completed at $(date)" >> /var/log/malware_scan.log
```

5.10 Make Executable

sudo chmod +x /usr/local/bin/weekly_malware_scan.sh

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo chmod +x /usr/local/bin/weekly_malware_scan.sh
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

5.11 Add to Crontab

sudo crontab -e

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo crontab -e
crontab: installing new crontab
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

Add this line:

0 3 * * 0 /usr/local/bin/weekly_malware_scan.sh

```
 GNU nano 8.7.1                                          /tmp/crontab.kfOZnX/crontab *
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 2 * * * /usr/local/bin/check_integrity.sh >> /var/log/integrity_check.log 2>&1
0 3 * * 0 /usr/local/bin/weekly_malware_scan.sh
```

Verify:

sudo crontab -l

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 2 * * * /usr/local/bin/check_integrity.sh >> /var/log/integrity_check.log 2>&1
0 3 * * 0 /usr/local/bin/weekly_malware_scan.sh
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

5.12 Test the Script Manually

sudo /usr/local/bin/weekly_malware_scan.sh

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo /usr/local/bin/weekly_malware_scan.sh
===== Weekly Malware Scan Report =====
Date: Sat Jul  4 08:19:25 PM PKT 2026
```

Check the log:

cat /var/log/malware_scan.log

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# cat /var/log/malware_scan.log
/home/abdul/.bash_history: OK
/home/abdul/.bashrc: OK
/home/abdul/.ssh/id_rsa.pub: OK
/home/abdul/.ssh/id_rsa: OK
/home/abdul/.bash_logout: OK
/home/abdul/.profile: OK

----------- SCAN SUMMARY -----------
Known viruses: 3627891
Engine version: 1.4.4
Scanned directories: 2
Scanned files: 6
Infected files: 0
Data scanned: 0.00 MB
Data read: 0.00 MB (ratio 0.00:1)
Time: 19.618 sec (0 m 19 s)
Start Date: 2026:07:04 20:19:25
End Date:   2026:07:04 20:19:45
/tmp/scan_test/eicar.com: Empty file
/tmp/scan_test/file2.txt: OK
/tmp/scan_test/file1.txt: OK
/tmp/important_data.txt: OK
/tmp/secure_file.txt: OK
/tmp/checksums.sha256: OK
/tmp/checksum.sha256: Empty file
/tmp/imp.txt: OK
/tmp/secure.txt: OK
/tmp/.X1-lock: OK
/tmp/.X0-lock: OK

----------- SCAN SUMMARY -----------
```

LAB 6 — File Monitoring and Event Auditing with AuditD

Executive summary

Installed and enabled AuditD, then configured watch rules to monitor changes to critical files (/etc/passwd, /etc/shadow, /etc/sudoers, sshd_config, /etc/group, /etc/gshadow). Triggered a test event by editing sudoers and confirmed detection via ausearch and raw audit logs. Generated summary audit reports (aureport) and built an automated audit reporting script — including failed login detection — scheduled daily at 4 AM via cron.

6.1 Install AuditD

sudo apt-get install auditd -y

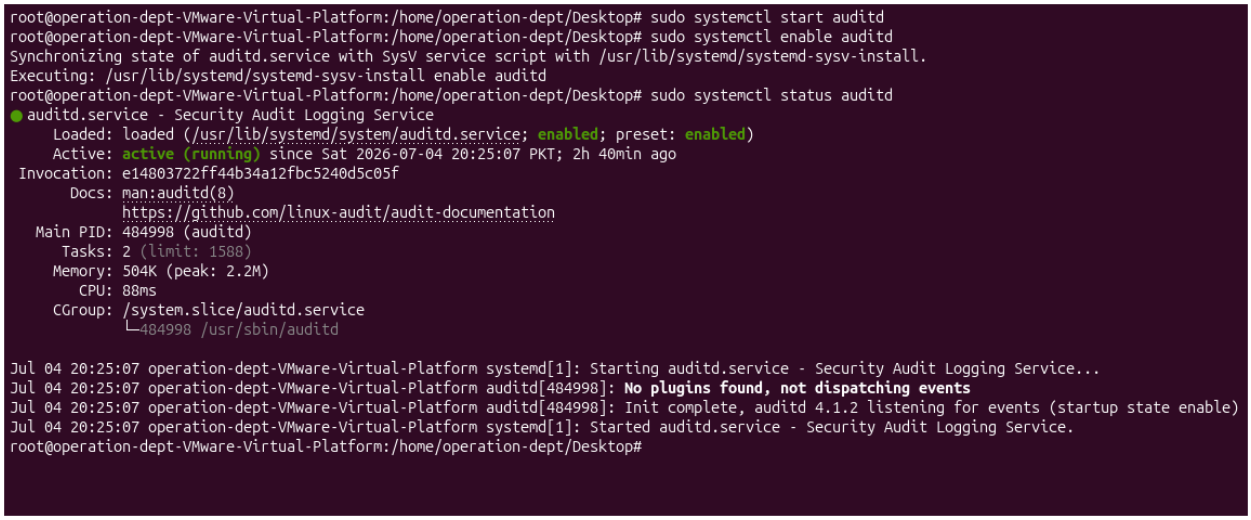
```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo apt-get install auditd -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following additional packages will be installed:
  libauparse0t64 libauplugin1
Suggested packages:
  audispd-plugins
The following NEW packages will be installed:
  auditd libauparse0t64 libauplugin1
0 upgraded, 3 newly installed, 0 to remove and 1 not upgraded.
Need to get 298 kB of archives.
After this operation, 975 kB of additional disk space will be used.
Get:1 http://pk.archive.ubuntu.com/ubuntu resolute/main amd64 libauparse0t64 amd64 1:4.1.2-1build1 [63.4 kB]
Get:2 http://pk.archive.ubuntu.com/ubuntu resolute/main amd64 libauplugin1 amd64 1:4.1.2-1build1 [9,978 B]
Get:3 http://pk.archive.ubuntu.com/ubuntu resolute/main amd64 auditd amd64 1:4.1.2-1build1 [224 kB]
Fetched 298 kB in 11s (27.4 kB/s)
Selecting previously unselected package libauparse0t64:amd64.
(Reading database… 158217 files and directories currently installed.)
Preparing to unpack …/libauparse0t64_1%3a4.1.2-1build1_amd64.deb…
Adding “diversion of /lib/x86_64-linux-gnu/libauparse.so.0 to /lib/x86_64-linux-gnu/libauparse.so.0.usr-is-merged by libauparse0t64”
Adding “diversion of /lib/x86_64-linux-gnu/libauparse.so.0.0.0 to /lib/x86_64-linux-gnu/libauparse.so.0.0.0.usr-is-merged by libauparse0t64”
Unpacking libauparse0t64:amd64 (1:4.1.2-1build1)…
Selecting previously unselected package libauplugin1:amd64.
Preparing to unpack …/libauplugin1_1%3a4.1.2-1build1_amd64.deb…
Unpacking libauplugin1:amd64 (1:4.1.2-1build1)…
Selecting previously unselected package auditd.
```

6.2 Start and Enable AuditD

sudo systemctl start auditd

sudo systemctl enable auditd

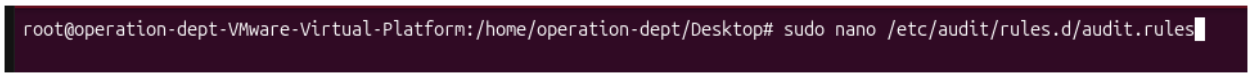
sudo systemctl status auditd



Expected: Active: active (running), log: "Init complete, auditd listening for events"

6.3 Add Audit Rules for File Monitoring

sudo nano /etc/audit/rules.d/audit.rules



Rule file content:

First rule - delete all

-D

Increase the buffers to survive stress events.

Make this bigger for busy systems

-b 8192

This determines how long to wait in burst of events

--backlog_wait_time 60000

Set failure mode to syslog

-f 1

-w /etc/passwd -p wa -k etc_passwd_changes

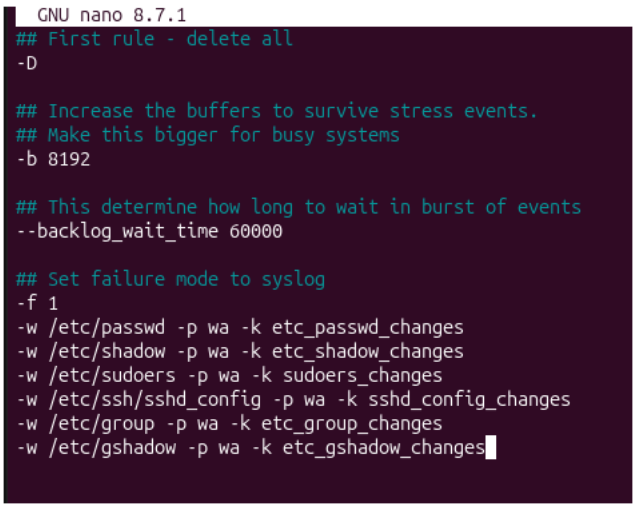
-w /etc/shadow -p wa -k etc_shadow_changes

-w /etc/sudoers -p wa -k sudoers_changes

-w /etc/ssh/sshd_config -p wa -k sshd_config_changes

-w /etc/group -p wa -k etc_group_changes

-w /etc/gshadow -p wa -k etc_gshadow_changes



6.4 Trigger a Test Audit Event

sudo nano /etc/sudoers

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo nano /etc/sudoers
```

At the end of the file, add a harmless comment line:

Trust audit rules

Then Save and exit

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo ausearch -k sudoers_changes
----
time->Sat Jul  4 23:08:24 2026
type=PROCTITLE msg=audit(1783188504.990:711): proctitle=6175646974636C002D77002F6574632F706173737764002D70007761002D6B006574635F
616E676573
type=PATH msg=audit(1783188504.990:711): item=0 name="/etc/" inode=655361 dev=08:02 mode=040755 ouid=0 ogid=0 rdev=00:00 nametype=
fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1783188504.990:711): cwd="/home/operation-dept/Desktop"
type=SOCKADDR msg=audit(1783188504.990:711): saddr=100000000000000000000000
type=SYSCALL msg=audit(1783188504.990:711): arch=c000003e syscall=44 success=yes exit=1088 a0=4 a1=7ffcc0c713d0 a2=440 a3=0 items=
9096 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts2 ses=3 comm="auditctl" exe="/usr/sbin/auditctl" sub
ll)
type=CONFIG_CHANGE msg=audit(1783188504.990:711): auid=1000 ses=3 subj=unconfined op=add_rule key="etc_sudoers_changes" list=4 res
----
time->Sat Jul  4 23:08:36 2026
type=PROCTITLE msg=audit(1783188516.290:717): proctitle=6175646974636C002D77002F6574632F706173737764002D70007761002D6B007375646F
73
type=PATH msg=audit(1783188516.290:717): item=0 name="/etc/" inode=655361 dev=08:02 mode=040755 ouid=0 ogid=0 rdev=00:00 nametype=
fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1783188516.290:717): cwd="/home/operation-dept/Desktop"
type=SOCKADDR msg=audit(1783188516.290:717): saddr=100000000000000000000000
type=SYSCALL msg=audit(1783188516.290:717): arch=c000003e syscall=44 success=yes exit=1084 a0=4 a1=7ffdc9948370 a2=43c a3=0 items=
9108 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts2 ses=3 comm="auditctl" exe="/usr/sbin/auditctl" sub
ll)
type=CONFIG_CHANGE msg=audit(1783188516.290:717): auid=1000 ses=3 subj=unconfined op=add_rule key="sudoers_changes" list=4 res=1
----
time->Sat Jul  4 23:10:42 2026
```

6.5 Search Audit Logs for Changes

sudo ausearch -k sudoers_changes

This shows detailed audit records (PROCTITLE, PATH, CWD, SOCKADDR, SYSCALL, CONFIG_CHANGE) for each time /etc/sudoers (and /etc/) was accessed, tagged with the key sudoers_changes / etc_sudoers_changes.

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo ausearch -k sudoers_changes
----
time->Sat Jul  4 23:08:24 2026
type=PROCTITLE msg=audit(1783188504.990:711): proctitle=617564697463746C002D77002F6574632F706173737764002D70007761002D6B006574635F7
616E676573
type=PATH msg=audit(1783188504.990:711): item=0 name="/etc/" inode=655361 dev=08:02 mode=040755 ouid=0 ogid=0 rdev=00:00 nametype=P
fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1783188504.990:711): cwd="/home/operation-dept/Desktop"
type=SOCKADDR msg=audit(1783188504.990:711): saddr=100000000000000000000000
type=SYSCALL msg=audit(1783188504.990:711): arch=c000003e syscall=44 success=yes exit=1088 a0=4 a1=7ffcc0c713d0 a2=440 a3=0 items=1
9096 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts2 ses=3 comm="auditctl" exe="/usr/sbin/auditctl" subj
ll)
type=CONFIG_CHANGE msg=audit(1783188504.990:711): auid=1000 ses=3 subj=unconfined op=add_rule key="etc_sudoers_changes" list=4 res=
----
time->Sat Jul  4 23:08:36 2026
type=PROCTITLE msg=audit(1783188516.290:717): proctitle=617564697463746C002D77002F6574632F706173737764002D70007761002D6B007375646F6
73
type=PATH msg=audit(1783188516.290:717): item=0 name="/etc/" inode=655361 dev=08:02 mode=040755 ouid=0 ogid=0 rdev=00:00 nametype=P
fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1783188516.290:717): cwd="/home/operation-dept/Desktop"
type=SOCKADDR msg=audit(1783188516.290:717): saddr=100000000000000000000000
type=SYSCALL msg=audit(1783188516.290:717): arch=c000003e syscall=44 success=yes exit=1084 a0=4 a1=7ffdc9948370 a2=43c a3=0 items=1
9108 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts2 ses=3 comm="auditctl" exe="/usr/sbin/auditctl" subj
ll)
type=CONFIG_CHANGE msg=audit(1783188516.290:717): auid=1000 ses=3 subj=unconfined op=add_rule key="sudoers_changes" list=4 res=1
----
time->Sat Jul  4 23:10:42 2026
type=PROCTITLE msg=audit(1783188642.972:732): proctitle=617564697463746C002D44
type=SOCKADDR msg=audit(1783188642.972:732): saddr=100000000000000000000000
type=SYSCALL msg=audit(1783188642.972:732): arch=c000003e syscall=44 success=yes exit=1088 a0=3 a1=7ffdc60fc710 a2=440 a3=0 items=0
```

6.6 View Detailed Raw Audit Logs

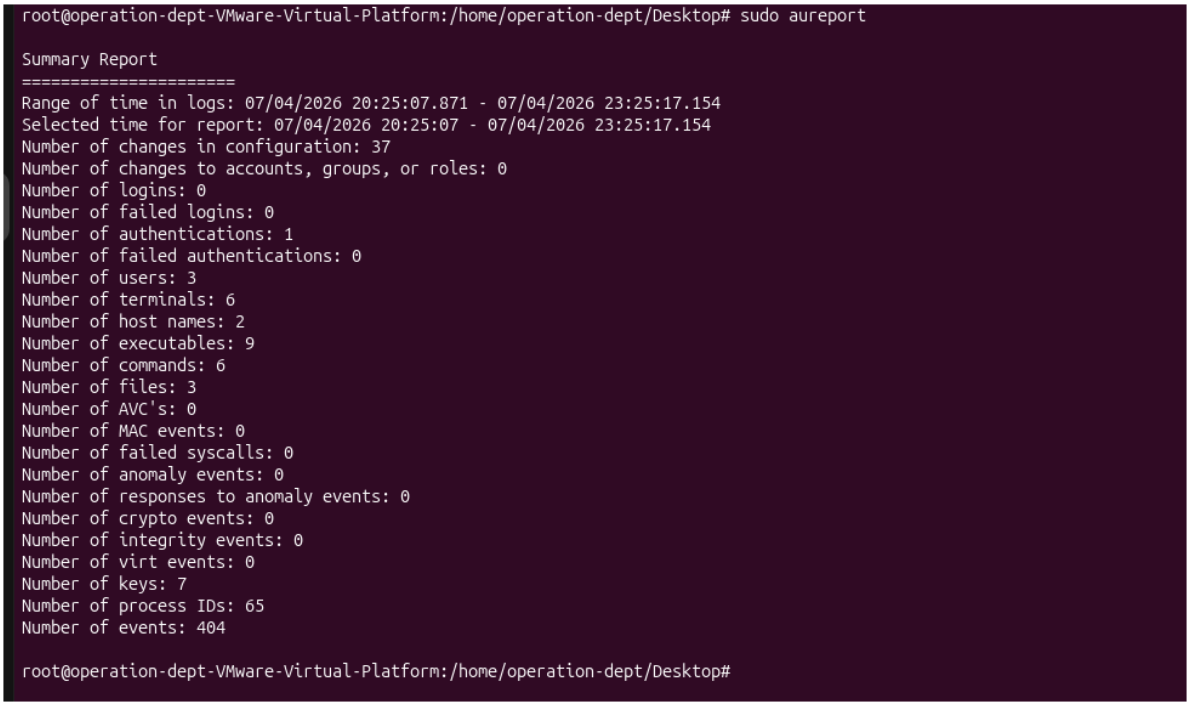
sudo tail -20 /var/log/audit/audit.log

Shows raw audit events: USER_END, USER_ACCT, USER_START, SYSCALL, PATH, PROCTITLE, etc., including sudo session opens/closes for user abdul/root.

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo tail -20 /var/log/audit/audit.log
type=USER_END msg=audit(1783189144.892:805): pid=489307 uid=0 auid=1000 ses=3 subj=unconfined msg='op=PAM:session_close grantors=pa
m_env,pam_permit,pam_umask,pam_unix acct="root" exe="/usr/lib/cargo/bin/sudo" hostname=operation-dept-VMware-Virtual-Platform addr=
/1 res=success'UID="root" AUID="operation-dept"
type=USER_ACCT msg=audit(1783189187.302:806): pid=489316 uid=0 auid=1000 ses=3 subj=unconfined msg='op=PAM:accounting grantors=pam_
r acct="root" exe="/usr/lib/cargo/bin/sudo" hostname=operation-dept-VMware-Virtual-Platform addr=? terminal=/dev/pts/1 res=success'
eration-dept"
type=USER_START msg=audit(1783189187.302:807): pid=489316 uid=0 auid=1000 ses=3 subj=unconfined msg='op=PAM:session_open grantors=p
am_env,pam_permit,pam_umask,pam_unix acct="root" exe="/usr/lib/cargo/bin/sudo" hostname=operation-dept-VMware-Virtual-Platform addr
s/1 res=success'UID="root" AUID="operation-dept"
type=SYSCALL msg=audit(1783189200.295:808): arch=c000003e syscall=257 success=yes exit=3 a0=ffffffffffffff9c a1=5c02e25ed860 a2=241
d=489318 pid=489319 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts2 ses=3 comm="nano" exe="/usr/bin/nano
ey="sudoers_changes"ARCH=x86_64 SYSCALL=openat AUID="operation-dept" UID="root" GID="root" EUID="root" SUID="root" FSUID="root" EGI
" FSGID="root"
type=CWD msg=audit(1783189200.295:808): cwd="/home/operation-dept/Desktop"
type=PATH msg=audit(1783189200.295:808): item=0 name="/etc/sudoers" inode=655814 dev=08:02 mode=0100440 ouid=0 ogid=0 rdev=00:00 na
p=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0OUID="root" OGID="root"
type=PROCTITLE msg=audit(1783189200.295:808): proctitle=6E616E6F002F6574632F7375646F657273
type=USER_END msg=audit(1783189200.557:809): pid=489316 uid=0 auid=1000 ses=3 subj=unconfined msg='op=PAM:session_close grantors=pa
m_env,pam_permit,pam_umask,pam_unix acct="root" exe="/usr/lib/cargo/bin/sudo" hostname=operation-dept-VMware-Virtual-Platform addr=
/1 res=success'UID="root" AUID="operation-dept"
type=USER_ACCT msg=audit(1783189212.121:810): pid=489327 uid=0 auid=1000 ses=3 subj=unconfined msg='op=PAM:accounting grantors=pam_
```

6.7 Generate a Summary Audit Report

sudo aureport



6.8 Create an Audit Monitoring Report Script

sudo nano /usr/local/bin/audit_report.sh

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo nano /usr/local/bin/audit_report.sh
```

Script content:

bash

#!/bin/bash

echo "===== AuditD Monitoring Report ====="

echo "Date: $(date)"

echo ""

Generate audit report

sudo aureport

Search for suspicious activities

echo ""

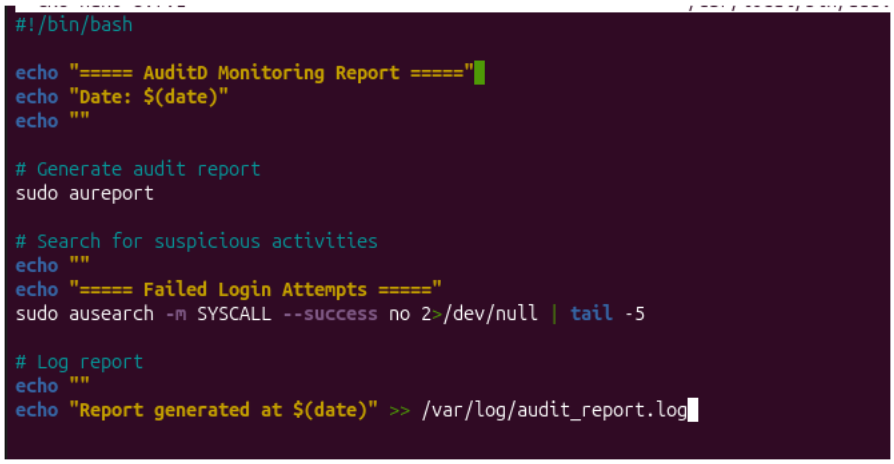
echo "===== Failed Login Attempts ====="

sudo ausearch -m SYSCALL --success no **2**>/dev/null | tail -5

Log report

echo ""

echo "Report generated at $(date)" >> /var/log/audit_report.



6.9 Make Executable and Verify by Running

sudo chmod +x /usr/local/bin/audit_report.sh

sudo /usr/local/bin/audit_report.sh

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo chmod +x /usr/local/bin/audit_report.sh
```

Verifying by run

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo /usr/local/bin/audit_report.sh
===== AuditD Monitoring Report =====
Date: Sat Jul  4 11:28:04 PM PKT 2026


Summary Report
======================
Range of time in logs: 07/04/2026 20:25:07.871 - 07/04/2026 23:28:04.208
Selected time for report: 07/04/2026 20:25:07 - 07/04/2026 23:28:04.208
Number of changes in configuration: 37
Number of changes to accounts, groups, or roles: 0
Number of logins: 0
Number of failed logins: 0
Number of authentications: 1
Number of failed authentications: 0
Number of users: 3
Number of terminals: 7
Number of host names: 2
Number of executables: 9
Number of commands: 6
Number of files: 3
Number of AVC's: 0
Number of MAC events: 0
Number of failed syscalls: 0
Number of anomaly events: 0
Number of responses to anomaly events: 0
Number of crypto events: 0
Number of integrity events: 0
Number of virt events: 0
Number of keys: 7
Number of process IDs: 69
Number of events: 415


===== Failed Login Attempts =====

root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

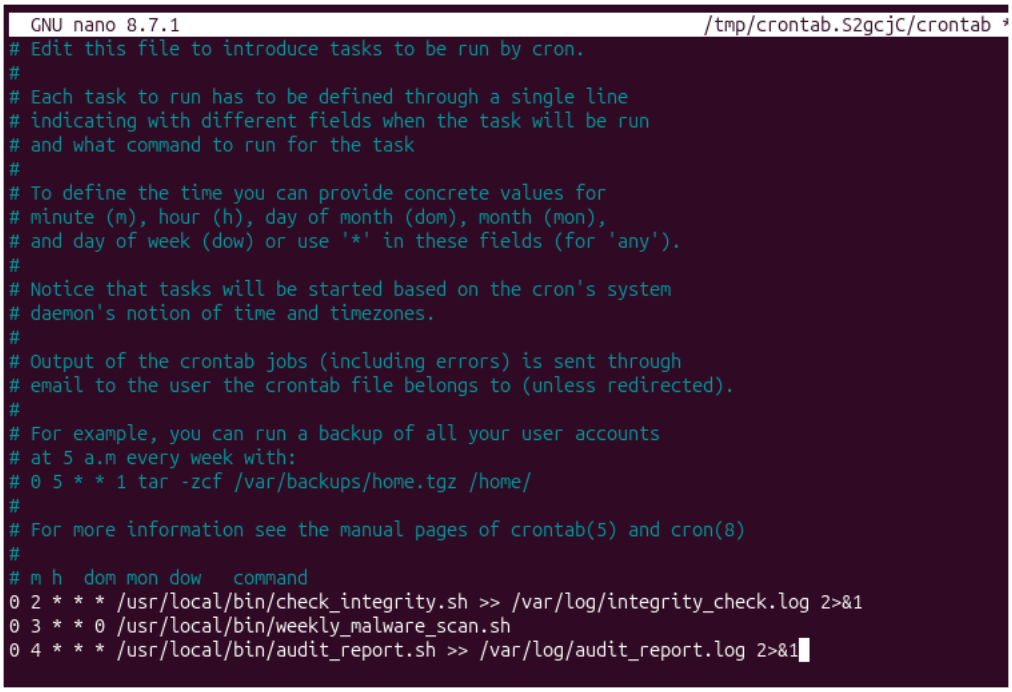
6.10 Add to Crontab

sudo crontab -e

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo crontab -e
crontab: installing new crontab
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

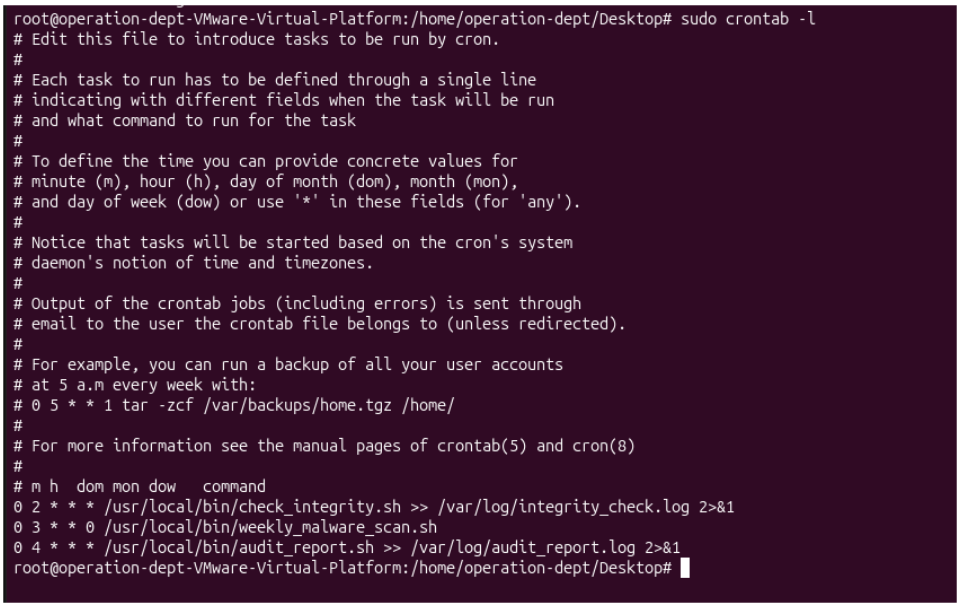
Add:

0 4 * * * /usr/local/bin/audit_report.sh >> /var/log/audit_report.log 2>&1



Verify final crontab list (should now include all previous jobs):

sudo crontab -l



LAB 7 — Kernel Management & System Cleanup

Executive summary

Reviewed installed kernels and kernel-related packages, then removed unused/unsigned kernel images and headers, freeing ~127–178 MB of disk space via autoremove and autoclean. Verified the active kernel (7.0.0-27-generic) remained intact and functional post-cleanup, resolved broken dependencies (apt-get check returned clean), and confirmed reduced boot partition usage. Built and scheduled an automated weekly system cleanup script (Sundays at 5 AM) that logs disk space before/after cleanup — completing a full set of four cron-scheduled maintenance jobs (integrity check, malware scan, audit report, system cleanup).

7.1 Check Current Running Kernel

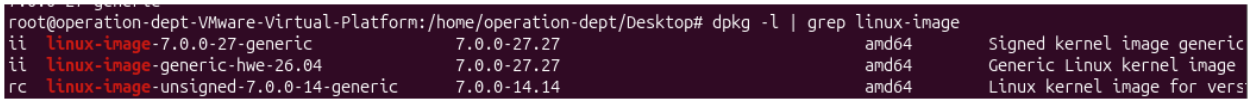
uname -r



Output: 7.0.0-27-generic

7.2 List All Installed Kernels

dpkg -l | grep linux-image



7.3 List All Kernel-Related Packages

apt list --installed | grep linux-

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# apt list --installed | grep linux-

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

binutils-x86-64-linux-gnu/resolute,now 2.46-3ubuntu2 amd64 [installed,automatic]
cpp-15-x86-64-linux-gnu/resolute,now 15.2.0-16ubuntu1 amd64 [installed,automatic]
cpp-x86-64-linux-gnu/resolute,now 4:15.2.0-5ubuntu1 amd64 [installed,automatic]
linux-base/resolute,now 4.15ubuntu5 all [installed,automatic]
linux-firmware-amd-graphics/resolute,now 20260319.git217ca6e4-0ubuntu3 all [installed,automatic]
linux-firmware-amd-misc/resolute,now 20260319.git217ca6e4-0ubuntu1 all [installed,automatic]
linux-firmware-broadcom-wireless/resolute,now 20260319.git217ca6e4-0ubuntu1 all [installed,automatic]
linux-firmware-intel-graphics/resolute,now 20260319.git217ca6e4-0ubuntu2 all [installed,automatic]
linux-firmware-intel-misc/resolute,now 20260319.git217ca6e4-0ubuntu1 all [installed,automatic]
linux-firmware-intel-wireless/resolute,now 20260319.git217ca6e4-0ubuntu2 all [installed,automatic]
linux-firmware-marvell-prestera/resolute,now 20260319.git217ca6e4-0ubuntu1 all [installed,automatic]
linux-firmware-marvell-wireless/resolute,now 20260319.git217ca6e4-0ubuntu1 all [installed,automatic]
linux-firmware-mediatek/resolute,now 20260319.git217ca6e4-0ubuntu1 all [installed,automatic]
linux-firmware-mellanox-spectrum/resolute,now 20260319.git217ca6e4-0ubuntu1 all [installed,automatic]
linux-firmware-misc/resolute,now 20260319.git217ca6e4-0ubuntu2 all [installed,automatic]
linux-firmware-netronome/resolute,now 20260319.git217ca6e4-0ubuntu1 all [installed,automatic]
linux-firmware-nvidia-graphics/resolute,now 20260319.git217ca6e4-0ubuntu1 all [installed,automatic]
linux-firmware-qlogic/resolute,now 20260319.git217ca6e4-0ubuntu1 all [installed,automatic]
linux-firmware-qualcomm-graphics/resolute,now 20260319.git217ca6e4-0ubuntu1 all [installed,automatic]
linux-firmware-qualcomm-misc/resolute,now 20260319.git217ca6e4-0ubuntu2 all [installed,automatic]
linux-firmware-qualcomm-wireless/resolute,now 20260319.git217ca6e4-0ubuntu1 all [installed,automatic]
linux-firmware-realtek/resolute,now 20260319.git217ca6e4-0ubuntu1 all [installed,automatic]
linux-firmware/resolute,now 20260319.git217ca6e4.1ubuntu all [installed,automatic]
linux-generic-hwe-24.04/resolute-updates,resolute-security,now 7.0.0-27.27 amd64 [installed]
linux-generic-hwe-26.04/resolute-updates,resolute-security,now 7.0.0-27.27 amd64 [installed,automatic]
linux-headers-7.0.0-27-generic/resolute-updates,resolute-security,now 7.0.0-27.27 amd64 [installed,automatic]
linux-headers-7.0.0-27/resolute-updates,resolute-security,now 7.0.0-27.27 all [installed,automatic]
linux-headers-generic-hwe-26.04/resolute-updates,resolute-security,now 7.0.0-27.27 amd64 [installed,automatic]
linux-image-7.0.0-27-generic/resolute-updates,resolute-security,now 7.0.0-27.27 amd64 [installed,automatic]
linux-image-generic-hwe-26.04/resolute-updates,resolute-security,now 7.0.0-27.27 amd64 [installed,automatic]
```

7.4 List Installed Kernel Versions

sudo apt list --installed | grep "linux-image"

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo apt list --installed | grep "linux-image"

WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

linux-image-7.0.0-27-generic/resolute-updates,resolute-security,now 7.0.0-27.27 amd64 [installed,automatic]
linux-image-generic-hwe-26.04/resolute-updates,resolute-security,now 7.0.0-27.27 amd64 [installed,automatic]
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

7.5 Check Boot Partition Usage

df -h /boot

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# df -h /boot
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda2        20G   11G  8.4G  55% /
```

7.6 List All Installed Linux Packages (Full)

dpkg -l | grep linux

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# dpkg -l | grep linux
ii  binutils-x86-64-linux-gnu                2.46-3ubuntu2                          amd64        GNU binary utilities, for x86-64-linux-gnu targ
et
ii  console-setup-linux                      1.237ubuntu3                           all          Linux specific part of console-setup
ii  cpp-15-x86-64-linux-gnu                  15.2.0-16ubuntu1                       amd64        GNU C preprocessor for x86_64-linux-gnu
ii  cpp-x86-64-linux-gnu                     4:15.2.0-5ubuntu1                      amd64        GNU C preprocessor (cpp) for the amd64 architec
ture
ii  libselinux1:amd64                        3.9-4build1                            amd64        SELinux runtime shared libraries
ii  libv4l-0t64:amd64                        1.32.0-2ubuntu1                        amd64        Collection of video4linux support libraries
ii  libv4lconvert0t64:amd64                  1.32.0-2ubuntu1                        amd64        Video4linux frame format conversion library
ii  linux-base                               4.15ubuntu5                            all          Linux image base package
ii  linux-firmware                           20260319.git217ca6e4.1ubuntu           all          Firmware for Linux kernel drivers (metapackage)
ii  linux-firmware-amd-graphics              20260319.git217ca6e4-0ubuntu3          all          Firmware for AMD/ATI graphics
ii  linux-firmware-amd-misc                  20260319.git217ca6e4-0ubuntu1          all          Firmware for AMD NPU accelerators
ii  linux-firmware-broadcom-wireless         20260319.git217ca6e4-0ubuntu1          all          Firmware for Broadcom and Cypress Wi-Fi and Blu
etooth adapters
ii  linux-firmware-intel-graphics            20260319.git217ca6e4-0ubuntu2          all          Firmware for Intel graphics and IPU and VSC pro
cessors
ii  linux-firmware-intel-misc                20260319.git217ca6e4-0ubuntu1          all          Firmware for miscellaneous Intel devices and ad
apters
ii  linux-firmware-intel-wireless            20260319.git217ca6e4-0ubuntu2          all          Firmware for Intel Wi-Fi and Bluetooth adapters
ii  linux-firmware-marvell-prestera          20260319.git217ca6e4-0ubuntu1          all          Firmware for Marvell Prestera ASIC devices
ii  linux-firmware-marvell-wireless          20260319.git217ca6e4-0ubuntu1          all          Firmware for Marvell and NXP Wi-Fi adapters
ii  linux-firmware-mediatek                  20260319.git217ca6e4-0ubuntu1          all          Firmware for Mediatek Wi-Fi, Bluetooth, and Eth
ernet adapters and SoCs
ii  linux-firmware-mellanox-spectrum         20260319.git217ca6e4-0ubuntu1          all          Firmware for Mellanox Spectrum switches
ii  linux-firmware-misc                      20260319.git217ca6e4-0ubuntu2          all          Firmware for miscellaneous devices and adapters
ii  linux-firmware-netronome                 20260319.git217ca6e4-0ubuntu1          all          Firmware for Netronome Ethernet adapters
ii  linux-firmware-nvidia-graphics           20260319.git217ca6e4-0ubuntu1          all          Firmware for Nvidia graphics
ii  linux-firmware-qlogic                    20260319.git217ca6e4-0ubuntu1          all          Firmware for QLogic SCSI, FC, and IB host bus a
nd Ethernet adapters
ii  linux-firmware-qualcomm-graphics         20260319.git217ca6e4-0ubuntu1          all          Firmware for Qualcomm graphics and video proces
sors
ii  linux-firmware-qualcomm-misc             20260319.git217ca6e4-0ubuntu2          all          Firmware for miscellaneous Qualcomm devices
ii  linux-firmware-qualcomm-wireless         20260319.git217ca6e4-0ubuntu1          all          Firmware for Qualcomm and Atheros Wi-Fi and Blu
etooth adapters
ii  linux-firmware-realtek                   20260319.git217ca6e4-0ubuntu1          all          Firmware for Realtek Wi-Fi, Bluetooth, Ethernet
```

7.7 Remove Old/Unused Kernel Packages

sudo apt-get remove --purge linux-image-unsigned-* -y

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo apt-get remove --purge linux-image-unsigned-* -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'linux-image-unsigned-7.0.0-1004-azure' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-1007-oracle' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-14-realtime' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-22-generic' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-1009-nvidia' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-1006-nvidia' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-27-generic' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-14-generic' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-1013-nvidia' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-15-generic' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-2012-nvidia-bos' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-1002-gke' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-1003-gcp' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-1004-aws' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-22-realtime' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-1005-oracle' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-1005-gcp' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-1006-aws' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-1005-ibm' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-1005-oem' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-1007-gcp' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-1008-aws' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-1007-ibm' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-1009-ibm' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-1008-oem' for glob 'linux-image-unsigned-*'
Note, selecting 'linux-image-unsigned-7.0.0-1004-azure-fde' for glob 'linux-image-unsigned-*'
```

Clean up headers:

sudo apt-get remove --purge linux-headers-* -y

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo apt-get remove --purge linux-headers-* -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'linux-headers-generic-hwe-24.04' for glob 'linux-headers-*'
Note, selecting 'linux-headers-oracle' for glob 'linux-headers-*'
Note, selecting 'linux-headers-nvidia-hwe-24.04' for glob 'linux-headers-*'
Note, selecting 'linux-headers-octeon' for glob 'linux-headers-*'
Note, selecting 'linux-headers-azure-fde' for glob 'linux-headers-*'
Note, selecting 'linux-headers-7.0.0-1002-gke' for glob 'linux-headers-*'
Note, selecting 'linux-headers-generic' for glob 'linux-headers-*'
Note, selecting 'linux-headers-s390x' for glob 'linux-headers-*'
```

```
After this operation, 127 MB disk space will be freed.
(Reading database… 158318 files and directories currently installed.)
Removing linux-generic-hwe-24.04 (7.0.0-27.27)…
Removing linux-generic-hwe-26.04 (7.0.0-27.27)…
Removing linux-headers-generic-hwe-26.04 (7.0.0-27.27)…
Removing linux-headers-7.0.0-27-generic (7.0.0-27.27)…
Removing linux-headers-7.0.0-27 (7.0.0-27.27)…
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

Autoremove unused dependencies:

sudo apt-get autoremove -y

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo apt-get autoremove -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following packages will be REMOVED:
  amd64-microcode bpfcc-tools bpftrace ieee-data intel-microcode iucode-tool libbpfcc libc-dev-bin libc6-dev libclang-cpp21 libcl
  linux-image-generic-hwe-26.04 linux-libc-dev manpages-dev python3-bpfcc python3-netaddr rpcsvc-proto thermald ubuntu-kernel-acc
0 upgraded, 0 newly installed, 19 to remove and 1 not upgraded.
After this operation, 178 MB disk space will be freed.
(Reading database… 125672 files and directories currently installed.)
Removing linux-image-generic-hwe-26.04 (7.0.0-27.27)…
Removing amd64-microcode (3.20251202.1ubuntu2)…
Removing bpfcc-tools (0.35.0+ds-1ubuntu2)…
Removing bpftrace (0.25.0-1ubuntu1)…
Removing python3-netaddr (1.3.0-1build1)…
Removing ieee-data (20240722build1)…
Removing intel-microcode (3.20260210.1ubuntu2)…
Removing iucode-tool (2.3.1-3build2)…
Removing python3-bpfcc (0.35.0+ds-1ubuntu2)…
Removing libbpfcc:amd64 (0.35.0+ds-1ubuntu2)…
Removing libc6-dev:amd64 (2.43-2ubuntu2)…
Removing libc-dev-bin (2.43-2ubuntu2)…
Removing libclang-cpp21 (1:21.1.8-6ubuntu1)…
Removing libclang1-21 (1:21.1.8-6ubuntu1)…
Removing linux-libc-dev:amd64 (7.0.0-27.27)…
Removing manpages-dev (6.17-1)…
Removing rpcsvc-proto (1.4.3-1build1)…
Removing thermald (2.5.11-0ubuntu1)…
```

Autoclean package cache:

sudo apt-get autoclean -y

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo apt-get autoclean -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

7.8 Verify Cleanup

dpkg -l | grep linux-image

df -h /boot

du -sh /var/cache/apt/

uname -r

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# dpkg -l | grep linux-image
ii  linux-image-7.0.0-27-generic             7.0.0-27.27                             amd64        Signed kernel image generic
```

Boot partition

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# df -h /boot
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda2        20G  9.7G  8.9G  53% /
```

Total disk space

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# du -sh /var/cache/apt/
20K     /var/cache/apt/
```

Current kernel

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# uname -r
7.0.0-27-generic
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

Remove unwanted broken dependencies

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo apt-get --fix-broken install -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

Expected: only the currently running kernel remains (linux-image-7.0.0-27-generic), boot partition usage reduced, and uname -r unchanged (still 7.0.0-27-generic — confirming the active kernel was not removed).

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo apt-get check
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

7.9 Fix Broken Dependencies

sudo apt-get --fix-broken install -y

sudo apt-get check

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo apt-get install -f -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo apt-get autoremove --purge -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo apt-get clean
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

7.10 Remove Broken Libraries / Final Cleanup Pass

sudo apt-get install -f -y

sudo apt-get autoremove --purge -y

sudo apt-get clean

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo apt-get check
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

7.11 Verify System Health and Disk Space

sudo apt-get check

du -sh /boot

du -sh /var/cache/apt/

du -sh /var/lib/apt/

df -h /

uname -r

ls -lah /boot/

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# du -sh /boot
72M     /boot
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# du -sh /var/cache/apt/
102M    /var/cache/apt/
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# du -sh /var/lib/apt/
152M    /var/lib/apt/
```

Show freed space

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# df -h /
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda2        20G  9.8G  8.8G  53% /
```

Check the kernel still working

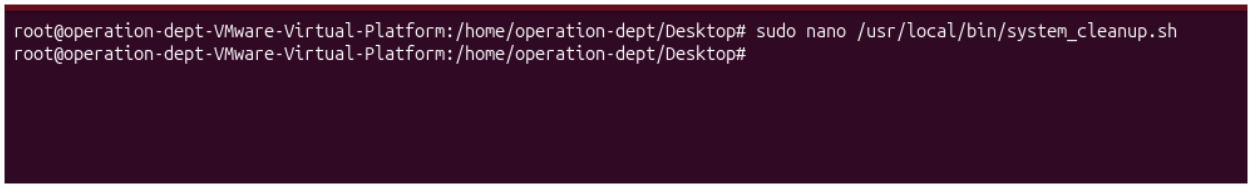
```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# uname -r
7.0.0-27-generic
```

Check boot files

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# ls -lah /boot/
total 64M
drwxr-xr-x  3 root root 4.0K Jul  4 23:47 .
drwxr-xr-x 20 root root 4.0K Jul  4 19:32 ..
-rw-------  1 root root  11M Jun 18 21:54 System.map-7.0.0-27-generic
-rw-r--r--  1 root root 302K Jun 18 21:54 config-7.0.0-27-generic
drwxr-xr-x  5 root root 4.0K Jul  3 22:35 grub
lrwxrwxrwx  1 root root   27 Jul  3 18:22 initrd.img -> initrd.img-7.0.0-27-generic
-rw-------  1 root root  36M Jul  4 23:47 initrd.img-7.0.0-27-generic
lrwxrwxrwx  1 root root   27 Jul  3 22:35 initrd.img.old -> initrd.img-7.0.0-27-generic
-rw-r--r--  1 root root 153K Nov 29  2025 mt86+ia32
-rw-r--r--  1 root root 158K Nov 29  2025 mt86+x64
lrwxrwxrwx  1 root root   24 Jul  3 18:22 vmlinuz -> vmlinuz-7.0.0-27-generic
-rw-------  1 root root  17M Jun 18 22:19 vmlinuz-7.0.0-27-generic
lrwxrwxrwx  1 root root   24 Jul  3 22:35 vmlinuz.old -> vmlinuz-7.0.0-27-generic
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

7.12 Create an Automated System Cleanup Script

sudo nano /usr/local/bin/system_cleanup.sh



Script content:

bash

#!/bin/bash

echo "===== System Cleanup Report ====="

echo "Date: $(date)"

echo ""

Record disk space before

echo "Disk space BEFORE cleanup:"

df -h / | tail -1

Remove old kernels

echo "Removing old kernel packages..."

sudo apt-get remove --purge linux-image-unsigned-* -y >> /dev/null **2>&1**

Clean dependencies

echo "Removing unused dependencies..."

sudo apt-get autoremove -y >> /dev/null **2>&1**

Clean package cache

echo "Cleaning package cache..."

sudo apt-get autoclean -y >> /dev/null **2>&1**

sudo apt-get clean -y >> /dev/null **2>&1**

Record disk space after

echo ""

echo "Disk space AFTER cleanup:"

df -h / | tail -1

Log results

echo "Cleanup completed at $(date)" >> /var/log/system_clea

```
 GNU nano 8.7.1                                  /usr/local/bin/system_clean
#!/bin/bash

echo "===== System Cleanup Report ====="
echo "Date: $(date)"
echo ""

# Record disk space before
echo "Disk space BEFORE cleanup:"
df -h / | tail -1

# Remove old kernels
echo "Removing old kernel packages..."
sudo apt-get remove --purge linux-image-unsigned-* -y >> /dev/null 2>&1

# Clean dependencies
echo "Removing unused dependencies..."
sudo apt-get autoremove -y >> /dev/null 2>&1

# Clean package cache
echo "Cleaning package cache..."
sudo apt-get autoclean -y >> /dev/null 2>&1
sudo apt-get clean -y >> /dev/null 2>&1

# Record disk space after
echo ""
echo "Disk space AFTER cleanup:"
df -h / | tail -1

# Log results
echo "Cleanup completed at $(date)" >> /var/log/system_cleanup.log
```

7.13 Make Executable and Test

sudo chmod +x /usr/local/bin/system_cleanup.sh

sudo /usr/local/bin/system_cleanup.sh

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo chmod +x /usr/local/bin/system_cleanup.sh
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo /usr/local/bin/system_cleanup.sh
===== System Cleanup Report =====
Date: Sun Jul  5 12:02:20 AM PKT 2026

Disk space BEFORE cleanup:
/dev/sda2        20G  9.8G  8.8G  53% /
Removing old kernel packages...
Removing unused dependencies...
Cleaning package cache...

Disk space AFTER cleanup:
/dev/sda2        20G  9.7G  8.9G  53% /
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

7.14 Schedule Weekly Automated Cleanup via Cron

sudo crontab -e

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo crontab -e
crontab: installing new crontab
```

Add:

0 5 * * 0 /usr/local/bin/system_cleanup.sh >> /var/log/system_cleanup.log 2>&1

```
 GNU nano 8.7.1                                          /tmp/crontab.NMvDgh/crontab *
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 2 * * * /usr/local/bin/check_integrity.sh >> /var/log/integrity_check.log 2>&1
0 3 * * 0 /usr/local/bin/weekly_malware_scan.sh
0 4 * * * /usr/local/bin/audit_report.sh >> /var/log/audit_report.log 2>&1
0 5 * * 0 /usr/local/bin/system_cleanup.sh >> /var/log/system_cleanup.log 2>&1
```

Verify final crontab (should now include all 4 scheduled jobs):

sudo crontab -l

```
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop# sudo crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command
0 2 * * * /usr/local/bin/check_integrity.sh >> /var/log/integrity_check.log 2>&1
0 3 * * 0 /usr/local/bin/weekly_malware_scan.sh
0 4 * * * /usr/local/bin/audit_report.sh >> /var/log/audit_report.log 2>&1
0 5 * * 0 /usr/local/bin/system_cleanup.sh >> /var/log/system_cleanup.log 2>&1
root@operation-dept-VMware-Virtual-Platform:/home/operation-dept/Desktop#
```

Conclusion

This series of labs delivered a complete, end-to-end hardened network and security operations environment built around a segmented VMware lab topology (host-only GREEN network and NAT-based RED network) protected by a Smoothwall firewall. Starting from foundational network isolation, the exercises progressively layered defense-in-depth controls onto the Ubuntu "Operation Dept" VM: hardened SSH and user access management, enforced strong password policies, continuous security auditing with Lynis, cryptographic file-integrity verification, malware detection with ClamAV, kernel-level event auditing with AuditD, and disciplined system/kernel housekeeping.

Beyond individual configuration steps, the labs emphasized **automation and sustainability** — every major control (integrity checks, malware scans, audit reporting, and system cleanup) was converted into a reusable script and scheduled via cron, ensuring these protections run continuously rather than as one-time exercises. This reflects real-world security operations practice, where hardening is not a single event but an ongoing, monitored process.

Collectively, the labs demonstrate practical competency across several core cybersecurity domains: network segmentation and firewall configuration, Linux system hardening, identity and access management, password policy enforcement, security auditing and compliance scoring, file integrity monitoring, malware defense, and change auditing. The consistent verification step at the end of each lab (checking logs, re-running scans, confirming service status) also reinforces a validate-after-configure mindset — a critical habit for anyone responsible for maintaining secure systems in production.

By the end of this exercise set, the lab environment represents a substantially more secure and auditable system than its initial state, with measurable improvements (e.g., Lynis hardening index increasing from 68 to 69) and a fully automated maintenance schedule in place to sustain that posture going forward.